# Contents

Prompt Response 6 entries


01 PROMPT
Detailed Syllabus Assignment No. Suggested List of Assignments 1 Apply conditional formatting using formulas to highlight all sales

02 RESPONSE
Excel Practical & Viva Exam Preparation Guide Your practical/oral exam is tomorrow, so the goal is not to memorize every Excel

03 PROMPT
explain me all the related concepts in all assignments with examples and how to recreate them in detail

04 RESPONSE
Complete Excel Practical & Viva Preparation — All Assignments Your exam is tomorrow, so we'll prepare you to do three things for

05 PROMPT
now also explain me all the excel concepts used in the assignment theory so I can properly give viva.

06 RESPONSE
Excel Theory for Practical & Viva Exam This guide focuses on the theory behind your Excel assignments: definitions, working

YOU

Detailed Syllabus Assignment No. Suggested List of Assignments 1 Apply conditional formatting using formulas to highlight all sales figures that exceed the average sales of the dataset. 2 Use a formula to highlight rows where the "Order Status" is "Pending" and the "Due Date" has passed. 3 Create a dependent dropdown list where selecting a "Category" automatically filters the available "Sub-category" options. 4 Implement data validation to ensure that dates entered fall within a specific range and numbers are within a specified limit. 5 Write a formula using IF, AND, and OR to categorize employees based on their performance score and attendance rate. 6 Create a nested IF formula to determine grade categories based on students' marks. 7 Extract the domain name from a list of email addresses using RIGHT, LEN, and SEARCH functions. Format names in "LASTNAME, Firstname" format using text functions 8 Modify an existing lookup formula to handle missing values using IFERROR and IFNA. 9 Use AVERAGEIF and COUNTIF to analyze a dataset containing employee salaries and departments. 10 Create a pivot table to analyze total sales by region and filter data for the top 3

Attached is my excel syllabus, the list of my assignments and the solved assignment files. tomorrow is my practical/oral exam of excel subject and I want to be properly prepared accordingly (primarily based on the assignments) because we may be asked to recreate a specific task from the assignments and give viva about it or excel in general. So help me prepare for it.

# Excel Practical & Viva Exam Preparation Guide

Your practical/oral exam is tomorrow, so the goal is not to memorize every Excel feature. You need to be able to **recreate the assigned tasks, write the correct formulas, explain why they work, and answer follow-up viva questions confidently.**

I'll organize your preparation into three priorities:

1. **Practical execution:** exact steps, formulas, cell references, and menu paths.
2. **Viva preparation:** definitions, differences, formula logic, and common follow-up questions.
3. **Mock practical:** tasks to solve independently so you can identify what still needs revision.

I've also considered your uploaded solved workbooks. They cover the four assignment groups: Excel essentials and conditional formatting, advanced functions and data analysis, PivotTables/charts/slicers/macros, and financial/business case studies. Your ten-item suggested list will receive the most detailed treatment, with additional topics from the solved files included where useful.

## 1. Your revision priorities

**Priority 1 — Must master**

Conditional formatting, IF/AND/OR, nested IF, text functions, IFERROR/IFNA, AVERAGEIF/COUNTIF, data validation, dependent dropdowns, and PivotTables.

**Priority 2 — Very likely useful**

Sorting, filtering, absolute references (`$A$1`), lookup functions, charts, PivotChart, slicers, date functions, and basic Excel terminology.

**Priority 3 — Broader syllabus**

Freeze Panes, Remove Duplicates, macros, AI-powered data types, and budget-versus-actual dashboards.

## 2. Excel fundamentals you must know

| Term | Meaning | Example |
|---|---|---|
| Workbook | An Excel file | `Sales.xlsx` |
| Worksheet | A sheet inside a workbook | `Sheet1` |
| Cell | Intersection of a row and column | `B5` |
| Range | A group of cells | `A2:A20` |
| Formula | An expression that calculates a result | `=A2+B2` |
| Function | A predefined calculation | `=SUM(A2:A20)` |
| Relative reference | Changes when copied | `A2` |
| Absolute reference | Row and column stay fixed when copied | `$A$2` |
| Mixed reference | Locks only the row or column | `$A2` or `A$2` |
| Criteria | A condition used for matching or filtering | `">1000"` |

Remember these rules:

- Formulas normally begin with `=`.
- Text values in formulas are enclosed in double quotation marks, such as `"Pending"`.
- `:` specifies a range, such as `A2:A10`.

- `&` joins text, such as `=A2&" "&B2`.
- `$` locks a reference when you copy a formula.
- Press **F4** while editing a cell reference to cycle through relative, absolute, and mixed references in standard desktop Excel.

A common viva question is: *What is the difference between a formula and a function?*

**Answer:** A formula is an expression created to calculate a result. A function is a predefined operation used within a formula. For example, `=A1+B1` is a formula, while `=SUM(A1:B1)` uses the SUM function.

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# Part A — Your 10 suggested assignments

## Assignment 1. Conditional formatting using formulas

**Task:** Highlight sales figures that exceed the average sales of the dataset.

Suppose sales are in cells `B2:B11`.

Example data

| Branch | Sales | Result |
|---|---|---|
| Branch A | ₹20,000 | Below average |
| Branch B | ₹40,000 | Above average |
| Branch C | ₹30,000 | At average |

Illustrative average: ₹30,000.

**Steps to perform in Excel**

1. Select `B2:B11`.
2. Go to **Home → Conditional Formatting → New Rule**.
3. Select **Use a formula to determine which cells to format**.
4. Enter the formula below.
5. Click **Format**, select a fill color, and click OK.

Formula:

```excel
=B2>AVERAGE($B$2:$B$11)
```

What does it mean?

- `B2` is the first cell being evaluated.
- `AVERAGE($B$2:$B$11)` calculates the average of all sales.
- `>` checks whether the current sale exceeds the average.
- `$B$2:$B$11` is absolute so the average range stays fixed as Excel evaluates each row.

To flag sales below 60% of the target, assuming targets are in column C:

```excel
=B2<60%*C2
```

If you need to highlight the **entire row** when its sales exceed the average, select the full table first, for example `A2:C11`, and use:

```excel
=$B2>AVERAGE($B$2:$B$11)
```

The dollar sign before B fixes the sales column, while the row number changes.

**Viva questions**

- **What is conditional formatting?** Formatting cells automatically when specified conditions are satisfied.
- **Why use a formula-based rule?** To apply flexible conditions that built-in rules may not support.
- **Does conditional formatting change the underlying value?** No. It changes the cell's appearance, not its stored value.
- **What is the difference between `>` and `>=`?** `>` excludes equality; `>=` includes it.

## Assignment 2. Highlight pending orders whose due dates have passed

Assume your columns are:

| Column | Field |
|---|---|
| A | Order ID |
| B | Customer Name |
| C | Order Status |
| D | Order Date |
| E | Due Date |

The task requires the order to satisfy **both** conditions: status is Pending and the due date is before today.

**Formula**

```excel
=AND($C2="Pending",$E2<TODAY())
```

**Steps**

1. Select the entire data range, such as `A2:E100`.
2. Go to **Home → Conditional Formatting → New Rule**.
3. Select **Use a formula to determine which cells to format**.
4. Enter the formula.
5. Choose a red fill and confirm.

`AND` returns TRUE only when every condition is TRUE. `TODAY()` returns the current date, so the rule updates as time passes.

Important distinction: if an order due today should also count as overdue, change `<TODAY()` to `<=TODAY()`. In the formula above, a due date equal to today is not considered overdue.

For a visual status system, you could use:

- Pending — yellow
- Delayed — red
- Completed — green

If the task asks for rows rather than individual cells, select the full table before creating the rule.

**Viva questions**

- What does `TODAY()` return? The current date, updated when Excel recalculates.
- Why use `AND`? Both the status and the overdue condition must be true.
- What happens if `TODAY()` is replaced with a fixed date? The rule no longer moves forward automatically.

## Assignment 3. Dependent dropdown list

**Task:** Selecting a category should restrict the available subcategories.

Example:

**Try the dependent-dropdown logic**

This is an illustration of the behavior; create the actual dropdowns using Excel's Data Validation menu.

**Category**

Electronics

**Available subcategories**

Laptop

Mobile

Keyboard

**How to create it in Excel**

Step 1: Create a source table on a separate sheet.

| A | B | C |
|---|---|---|
| Electronics | Stationery | Furniture |
| Laptop | Notebook | Chair |
| Mobile | Pen | Table |
| Keyboard | Marker | Desk |

Step 2: Create named ranges.

1. Select the Electronics subcategory cells, excluding the heading.
2. Click the **Name Box** to the left of the formula bar.
3. Type `Electronics` and press Enter.
4. Repeat for the other categories, naming the ranges `Stationery` and `Furniture`.

Step 3: Create the Category dropdown.

1. Select the category input cell, for example `E2`.
2. Go to **Data → Data Validation**.
3. Set Allow to **List**.
4. Set Source to the category heading range, for example:

```excel
=$A$1:$C$1
```

Step 4: Create the dependent Sub-category dropdown in `F2`.

1. Select `F2`.
2. Open **Data → Data Validation → List**.
3. Set Source to:

```excel
=INDIRECT(E2)
```

When `E2` contains `Electronics`, Excel uses the named range `Electronics` as the source of the second dropdown.

**Viva questions**

- **What is data validation?** A feature that restricts or controls the values entered into cells.
- **What is a dependent dropdown?** A dropdown whose available choices depend on another cell's selection.
- **Why use `INDIRECT` here?** It interprets the text in the category cell as a reference to a named range.
- **What is a limitation of this method?** Category names must match the named ranges. Spaces and special characters require appropriate naming or a conversion formula.

## Assignment 4. Data validation for dates and numbers

**Task:** Restrict dates to a specified period and numbers to a permitted limit.

### Date validation

Suppose the allowed dates are from 1 April 2026 through 31 March 2027.

1. Select the date input cells.
2. Go to **Data → Data Validation**.
3. Set Allow to **Date**.
4. Set Data to **between**.
5. Enter the start date and end date.

If you want a formula-based rule instead, assuming the input is in `A2`:

```excel
=AND(A2>=DATE(2026,4,1),A2<=DATE(2027,3,31))
```

### Number validation

To allow only whole numbers from 1 to 100:

1. Select the required cells.
2. Go to **Data → Data Validation**.
3. Set Allow to **Whole number**.
4. Set Data to **between**.
5. Enter 1 and 100.

A custom formula equivalent is:

```excel
=AND(A2>=1,A2<=100,MOD(A2,1)=0)
```

`MOD(A2,1)=0` ensures that the number has no fractional part.

**Viva questions**

- What is an input message? A message displayed when the user selects the cell.
- What is an error alert? A warning or restriction shown when an invalid value is entered.
- Does data validation automatically clean old invalid data? No. Existing entries may need to be checked separately.
- What is the difference between Whole number and Decimal validation? Whole number permits integers; Decimal permits fractional numeric values.

## Assignment 5. IF, AND and OR for employee classification

**Task:** Classify employees based on performance score and attendance.

The exact thresholds depend on the question. For practice, assume:

- Outstanding: score at least 90 **and** attendance at least 95%.
- Satisfactory: score at least 70 **or** attendance at least 85%.
- Otherwise: Needs Improvement.

Suppose performance is in `B2` and attendance is in `C2`.

```excel
=IF(AND(B2>=90,C2>=95%),"Outstanding",
   IF(OR(B2>=70,C2>=85%),"Satisfactory",
      "Needs Improvement"))
```

If Excel rejects a pasted multiline formula, enter it as one continuous line.

**Understand the logic**

- `IF(condition, result_if_true, result_if_false)` chooses between outcomes.
- `AND(condition1, condition2)` requires every condition to be true.
- `OR(condition1, condition2)` requires at least one condition to be true.
- The outer IF checks Outstanding first. Only if that fails does Excel test Satisfactory.

**Important:** The thresholds and rules above are illustrative, not universal. In your exam, use the exact criteria stated in the question.

**Viva questions**

- What is a logical function?
- What is the difference between AND and OR?
- Why do we use nested IF?
- What happens when none of the conditions is true?

## Assignment 6. Nested IF for student grades

Suppose marks out of 100 are in cell `B2`, and the required grade scale is:

| Marks | Grade |
|---|---|
| 90–100 | O |
| 80–89 | A+ |
| 70–79 | A |
| 60–69 | B+ |
| 50–59 | B |
| 40–49 | C |
| Below 40 | P |

For this example, use:

```excel
=IF(B2>=90,"O",
 IF(B2>=80,"A+",
 IF(B2>=70,"A",
 IF(B2>=60,"B+",
 IF(B2>=50,"B",
 IF(B2>=40,"C","P"))))))
```

This formula checks the highest threshold first. A mark of 85 fails `B2>=90`, but satisfies `B2>=80`, so the grade is A+.

**Common mistake:** Testing the lowest threshold first can produce incorrect results. If the first condition is `B2>=40`, a mark of 95 would receive C immediately.

If marks outside 0–100 must be rejected, use data validation or add a validity check to the formula. Also, use the exact grading scale specified by your examiner.

**Viva questions**

- What is a nested IF? An IF function placed inside another IF function.
- Why is condition order important? Excel returns the result associated with the first TRUE condition.
- What is an alternative to deeply nested IFs? `IFS` for ordered conditions, or a lookup table for more maintainable grading rules.

## Assignment 7. Text functions: domain extraction and name formatting

This assignment has two separate tasks.

### A. Extract the domain from an email address

Suppose cell `A2` contains:

```
krishna.patil@gmail.com
```

Expected output: `gmail.com`

Formula using the requested functions:

```excel
=RIGHT(A2,LEN(A2)-SEARCH("@",A2))
```

How it works:

1. `SEARCH("@",A2)` finds the position of the @ symbol.
2. `LEN(A2)` counts the total characters.
3. Subtracting the @ position from the total length gives the number of characters after @.
4. `RIGHT` extracts that many characters from the right.

For example, if the email is `krishna.patil@gmail.com`, the domain extracted is `gmail.com`.

### B. Format a name as `LASTNAME, Firstname`

Suppose `B2` contains `Krishna Patil`.

Formula:

```excel
=UPPER(RIGHT(B2,LEN(B2)-SEARCH(" ",B2)))
 & ", " &
 LEFT(B2,SEARCH(" ",B2)-1)
```

Expected output:

```
PATIL, Krishna
```

- `SEARCH(" ",B2)` finds the first space.
- `LEFT` extracts the first name.
- `RIGHT` extracts the surname.
- `UPPER` capitalizes the surname.
- `&` joins the extracted text with a comma and a space.

This formula assumes the name contains exactly a first name and a surname separated by a space. Multiple-word names need a more robust approach.

**Other text functions worth revising**

| Function | Purpose | Example |
|---|---|---|
| `LEFT` | Extract characters from the beginning | `=LEFT(A2,3)` |
| `RIGHT` | Extract characters from the end | `=RIGHT(A2,4)` |
| `MID` | Extract characters from a specified position | `=MID(A2,3,5)` |
| `LEN` | Count characters | `=LEN(A2)` |
| `SEARCH` | Find text, case-insensitively | `=SEARCH("@",A2)` |
| `FIND` | Find text, case-sensitively | `=FIND("@",A2)` |
| `SUBSTITUTE` | Replace matching text | `=SUBSTITUTE(A2,"/","-")` |
| `UPPER` | Convert to uppercase | `=UPPER(A2)` |

**Viva questions**

- Difference between FIND and SEARCH? FIND is case-sensitive; SEARCH is not.
- Difference between LEFT and MID? LEFT starts at the first character; MID starts at a specified position.
- What does LEN count? Characters, including spaces and punctuation.

## Assignment 8. IFERROR and IFNA for lookup formulas

**Task:** Display a meaningful message when a lookup cannot find a matching value.

Suppose a product code is entered in `E2`. The lookup table is `A2:C100`, where column A contains product codes and column B contains product names.

Using VLOOKUP:

```excel
=IFERROR(VLOOKUP(E2,$A$2:$C$100,2,FALSE),"Not Found")
```

This searches for the code in the first column of the table and returns the matching value from the second column. If an error occurs, it displays `"Not Found"`.

Using IFNA:

```excel
=IFNA(VLOOKUP(E2,$A$2:$C$100,2,FALSE),"Not Found")
```

**The difference is important:**

- `IFERROR` handles many Excel errors, including `#N/A`, `#REF!`, and `#VALUE!`.
- `IFNA` handles only `#N/A`.

Use `IFNA` when you want to handle missing lookup results without concealing unrelated errors. Use `IFERROR` when a broader fallback is appropriate.

**Viva questions**

- What does VLOOKUP do? Searches vertically in the first column of a table and returns a value from another column in the same row.
- What does `FALSE` mean in VLOOKUP? Exact match.
- What is `#N/A`? A value is not available, often because a lookup did not find a match.
- What is a limitation of VLOOKUP? It generally searches the first column and returns values from columns to its right.

## Assignment 9. AVERAGEIF and COUNTIF

**Task:** Analyze employee salaries by department and threshold.

Suppose:

- Column A: Employee Name
- Column B: Department
- Column C: Salary

### A. Average salary of the IT department

```excel
=AVERAGEIF(B2:B100,"IT",C2:C100)
```

Syntax:

```
AVERAGEIF(criteria_range, criteria, average_range)
```

Excel checks the department column and averages only salaries associated with IT.

### B. Count employees earning more than ₹50,000

```excel
=COUNTIF(C2:C100,">50000")
```

Syntax:

```
COUNTIF(range, criteria)
```

This counts salaries strictly greater than 50,000. To include 50,000, use `">=50000"`.

### C. Total salary for a selected department

Although not one of the two named functions in the question, `SUMIF` is also important:

```excel
=SUMIF(B2:B100,"IT",C2:C100)
```

It adds salaries only for employees in IT.

#### Viva questions

- What is the difference between COUNT and COUNTIF? COUNT counts numeric cells; COUNTIF counts cells meeting a specified condition.
- What is the difference between AVERAGE and AVERAGEIF? AVERAGE considers the supplied numeric values; AVERAGEIF averages only values meeting a criterion.
- Why put `">50000"` in quotes? It is a criterion expression containing a comparison operator.
- What does SUMIF do? Adds values that meet a specified condition.

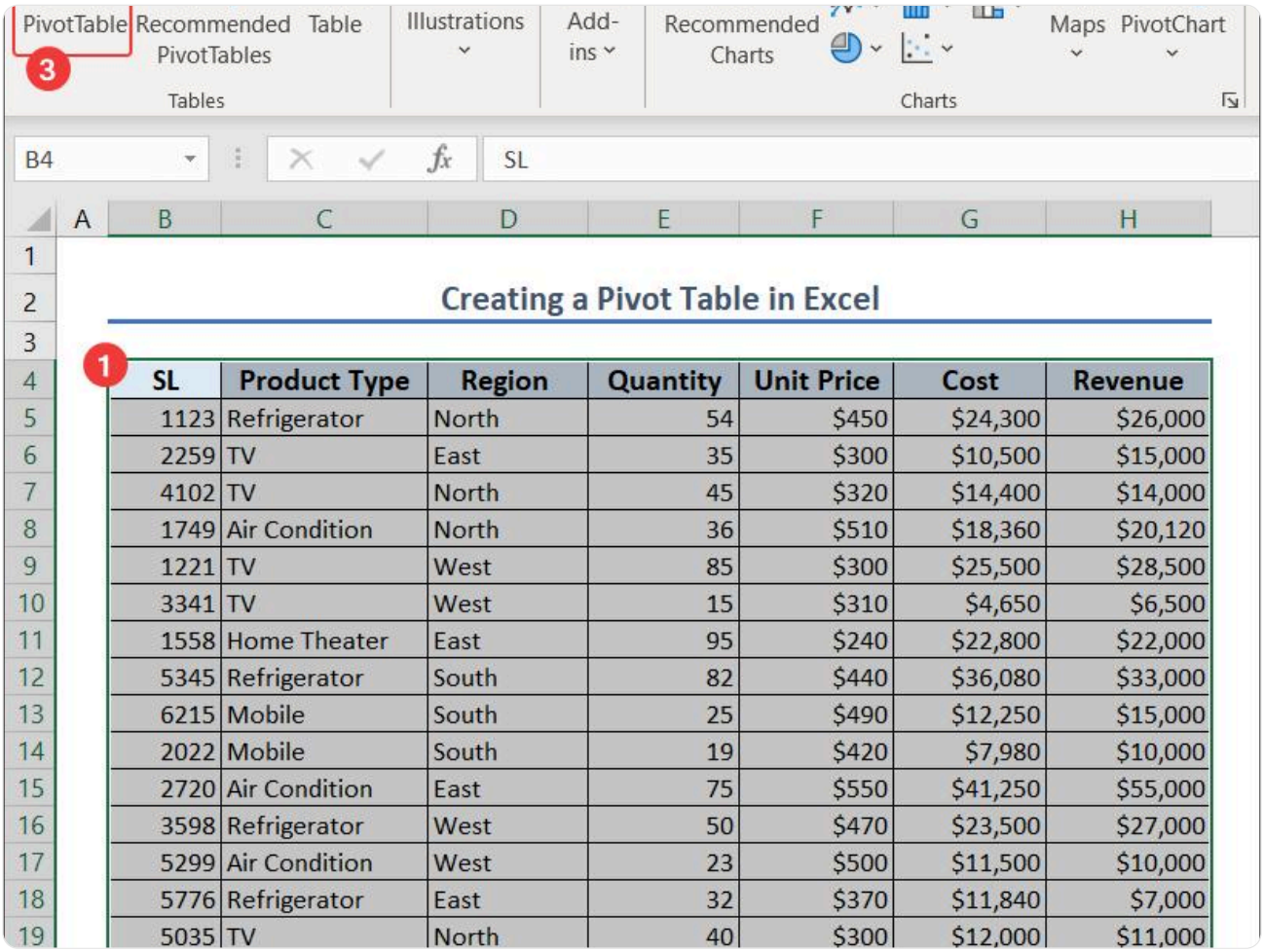
## Assignment 10. PivotTable for total sales by region and Top 3 regions

This is one of the most important practical tasks because it tests data preparation, summarization, filtering, and interpretation.

Suppose the source data contains:

| Date | Region | Product | Quantity | Sales Amount |
|---|---|---|---|---|
| 01-Oct | West | Laptop | 5 | 250000 |
| 02-Oct | North | Mobile | 10 | 180000 |
| 03-Oct | West | Mobile | 8 | 144000 |
| 04-Oct | South | Laptop | 4 | 200000 |

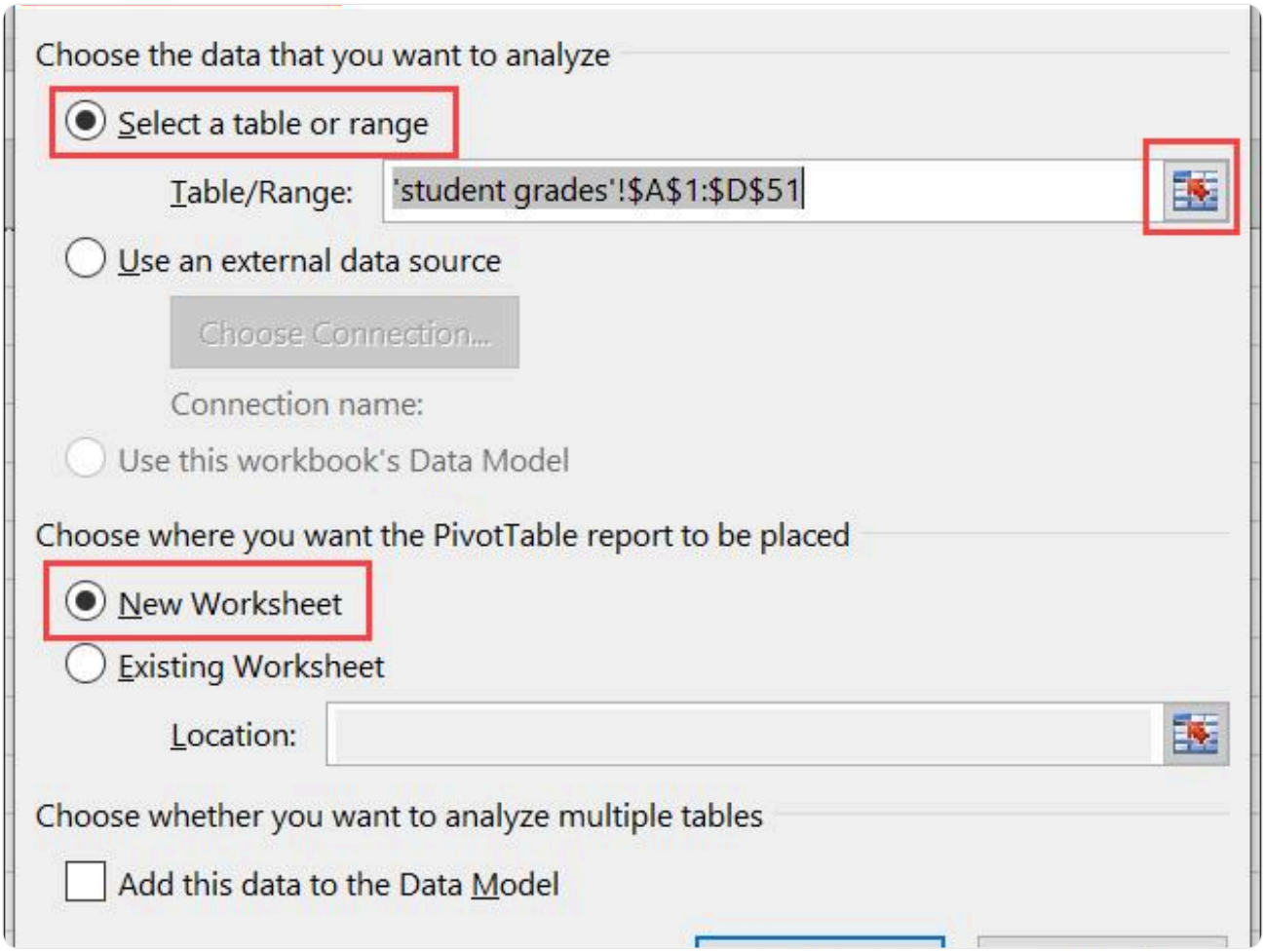
## Steps to create the PivotTable




**Step 1 — Select the source data**

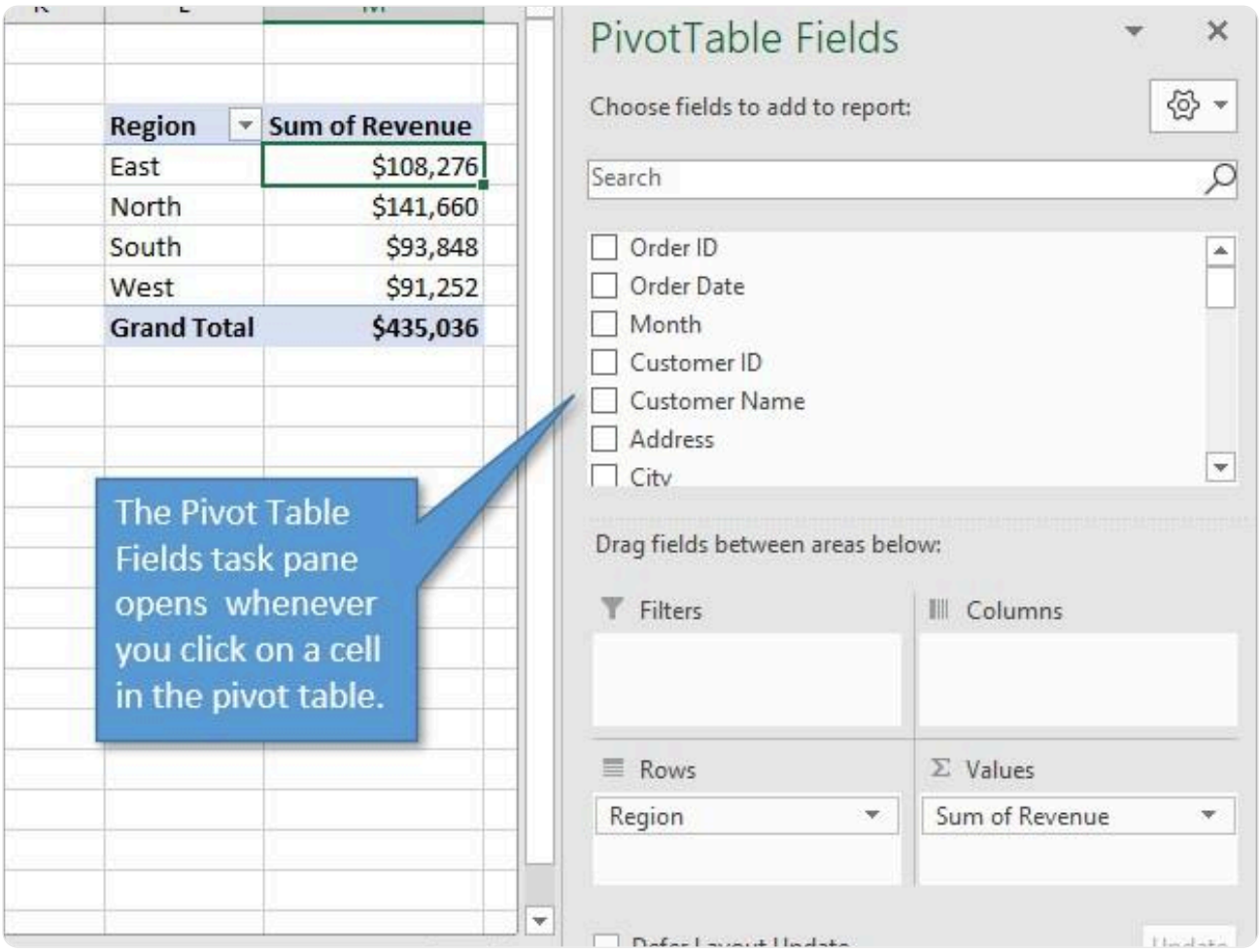
Click any cell inside the dataset. Ideally, format it as a table with **Ctrl+T** and ensure the first row contains unique

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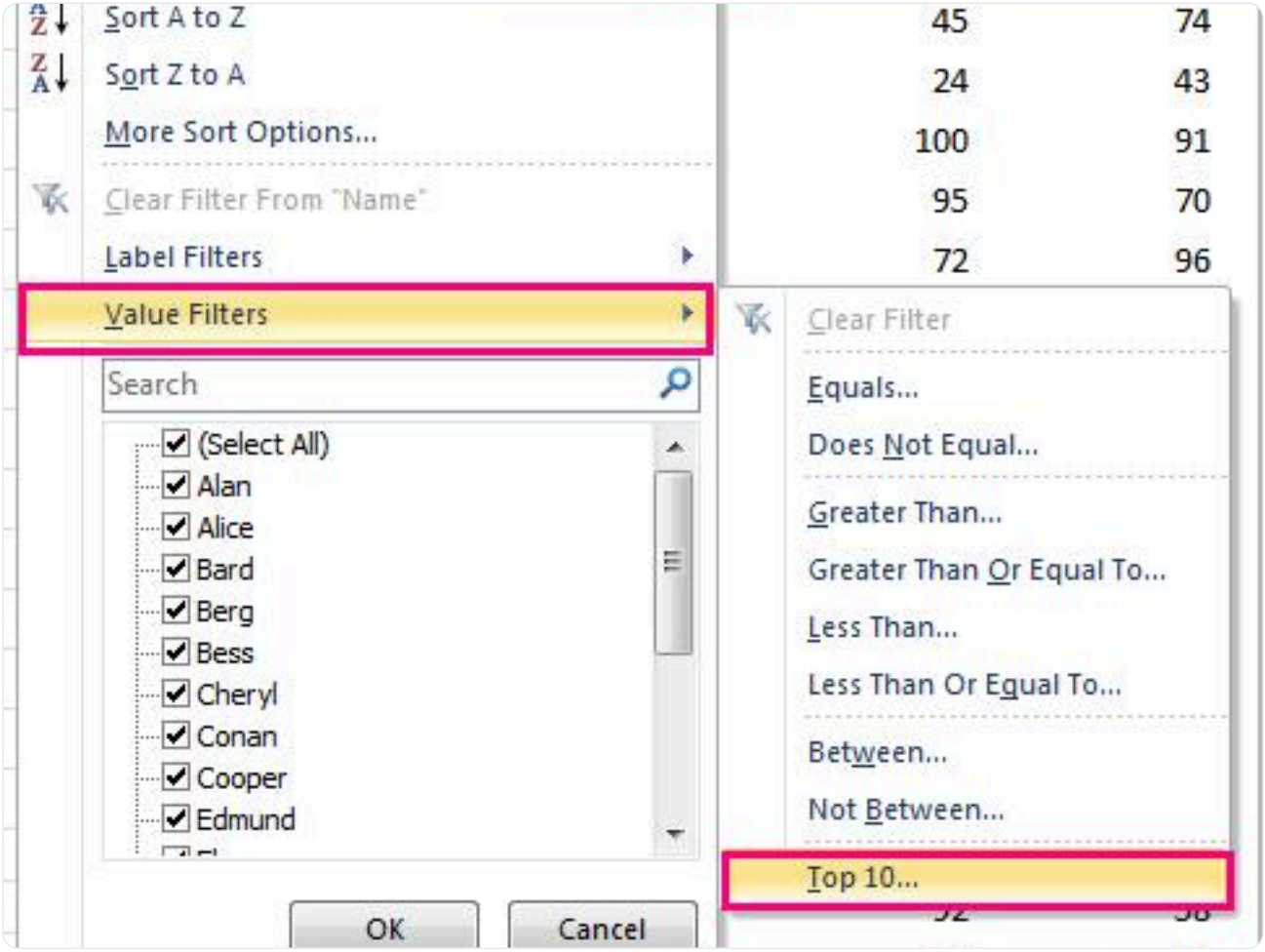

**Step 2 — Insert the Pivot-Table**

Go to **Insert → Pivot-Table**, check the source range, choose New Work-sheet, and click OK.




**Step 3 — Arrange the fields**

Drag `Region` to Rows and `Sales Amount` to Values. Ensure the calculation is **Sum of Sales Amount**, not Count.




**Step 4 — Filter the Top 3**

Open the drop-down beside the row labels. Choose **Value Filters → Top 10**, change 10 to 3, select Items and Sum of Sales Amount, then confirm.

The menu wording may differ slightly between Excel versions.

**Viva questions**

- What is a PivotTable? An interactive tool that summarizes and analyzes large datasets without requiring separate formulas for every group.
- What are Rows, Columns, Values, and Filters? They define how the data is grouped, compared, aggregated, and restricted.
- What is a PivotChart? A chart linked to a PivotTable that visualizes its summary.
- What is a slicer? A visual filtering control for PivotTables and compatible tables.
- Why might a PivotTable show Count instead of Sum? Excel may have detected text or mixed data types in the source field.
- How do you update a PivotTable after source data changes? Right-click it and select **Refresh**, or use **Data → Refresh All**.

---

# Part B — Additional topics from your solved workbooks

Your syllabus goes beyond the ten suggested questions. These topics are especially worth preparing because they appear in your uploaded Assignment 1–4 workbooks.

## 3. More conditional formatting and data management

### Highlight branches based on target achievement

Suppose actual sales are in `B2` and target sales are in `C2`.

| Condition | Formula | Format |
|---|---|---|
| More than 110% of target | `=B2>110%*C2` | Green |
| Between 90% and 110%, inclusive | `=AND(B2>=90%*C2,B2<=110%*C2)` | Yellow |
| Below 90% | `=B2<90%*C2` | Red |

Apply each rule to the relevant range. If the task asks you to color entire rows, select the whole table and anchor the appropriate column with `$`.

### Sort and filter

- **Sort:** Rearranges records by a field, such as salary from highest to lowest.
- **Filter:** Displays only records meeting selected conditions without deleting the other records.
- **Multi-level sort:** Use **Data → Sort → Add Level** to sort by program, then city, then score.

- **Remove Duplicates:** Use **Data → Remove Duplicates** and select the column(s) that determine whether a record is a duplicate. This operation deletes duplicate records from the selected data, so work on a copy when necessary.
- **Freeze Panes:** Use **View → Freeze Panes** to keep headers or key columns visible while scrolling. To freeze the top row, use **View → Freeze Top Row**.

Viva distinction: **sorting changes order; filtering changes visibility; Remove Duplicates removes records.**

## 4. More logical and lookup functions

These are present in the advanced-functions workbook.

| Function | What it does | Example |
|---|---|---|
| `VLOOKUP` | Looks down the first column of a table | `=VLOOKUP(E2,A2:C100,2,FALSE)` |
| `INDEX` | Returns a value at a position in a range | `=INDEX(B2:B100,5)` |
| `MATCH` | Finds a value's position | `=MATCH(E2,A2:A100,0)` |
| `INDEX` + `MATCH` | Finds a position and retrieves a related value | `=INDEX(B2:B100,MATCH(E2,A2:A100,0))` |
| `XLOOKUP` | Searches a range and returns the corresponding result | `=XLOOKUP(E2,A2:A100,B2:B100,"Not Found")` |
| `SUMIF` | Adds values meeting a condition | `=SUMIF(A2:A100,"IT",C2:C100)` |
| `MAX` | Returns the largest value | `=MAX(C2:C100)` |
| `ISNUMBER` | Checks whether a value is numeric | `=ISNUMBER(C2)` |

For `INDEX` + `MATCH`, the `0` in MATCH requests an exact match. Unlike a basic VLOOKUP, INDEX + MATCH can return values from a column to the left of the lookup column, provided the ranges are arranged accordingly.

### Date functions

Your workbook also includes project deadline and employee tenure analysis.

| Function | Purpose | Example |
|---|---|---|
| `TODAY()` | Current date | `=TODAY()` |
| `DATE(y,m,d)` | Creates a date | `=DATE(2026,10,27)` |
| `DATEDIF` | Calculates difference between dates | `=DATEDIF(A2,TODAY(),"y")` |
| `EOMONTH` | Returns the last day of a month | `=EOMONTH(A2,0)` |

For remaining days until a deadline in `B2`:

```excel
=B2-TODAY()
```

For a warning when a deadline is between today and seven days from today:

```excel
=AND(B2>=TODAY(),B2<=TODAY()+7)
```

A negative number of remaining days indicates that the deadline has passed.

## 5. PivotCharts, slicers, and macros

These topics feature in Assignment 3.

**PivotChart:** A chart based on a PivotTable. Use it to compare regions, product categories, monthly trends, or student attendance.

**Slicer:** A set of clickable buttons that filters a table or compatible PivotTable. Insert one from the table or PivotTable's contextual tab using **Insert Slicer** or **PivotTable Analyze → Insert Slicer**, depending on the Excel version.

**Macro:** A recorded or programmed sequence of actions used to automate repetitive tasks.

Basic macro workflow:

1. Enable the Developer tab if necessary.
2. Click **Developer → Record Macro**.
3. Enter a macro name and begin recording.
4. Perform the required formatting or reporting actions.
5. Stop recording.
6. Run the macro to repeat the actions.

Viva questions:

- What is VBA? Visual Basic for Applications, the programming language used to automate tasks in desktop Microsoft Office applications.
- What is the difference between a formula and a macro? A formula calculates a result in a cell; a macro automates a sequence of actions.
- Why is `.xlsm` used? It is an Excel Macro-Enabled Workbook format that can store VBA macros.
- What is a dashboard? A consolidated visual summary of important metrics, often using charts, tables, and key performance indicators.

**AI-powered data types:** Your final workbook includes a geography-enrichment case. In supported Excel versions and accounts, a city name can be converted through **Data → Data Types →**

**Geography** to access linked information. Availability depends on the Excel version, account, and connectivity; it is not guaranteed in every installation.

### 6. Financial case-study formulas

Your Assignment 4 workbook compares actual values with budgets.

Suppose budgeted expenditure is in `B2` and actual expenditure is in `C2`.

```excel
=C2-B2
```

This calculates variance. For expenditure, a positive variance means overspending.

```excel
=(C2-B2)/B2
```

This calculates variance percentage, assuming the budget is nonzero.

```excel
=IF(C2>B2,"Over Budget","Within Budget")
```

This classifies the result.

For income, however, actual income greater than budget is usually favorable. **Always interpret the sign of the variance according to the type of metric.**

Other useful formulas from the workbook include:

```excel
=SUM(B2:B13)
=AVERAGE(B2:B13)
=COUNTIF(C2:C100,"Critical")
=IFERROR(C2/B2,0)
```

The last formula substitutes zero when the division produces an error. In a real report, a message such as `"Check Budget"` may be more meaningful than zero.

---

## Part C — The viva question bank

Practice saying the answers aloud in one or two sentences. The examiner may ask you to explain your formula, change a condition, or demonstrate the result.

Viva flashcards **1 of 20**

### What is Microsoft Excel?

Try answering aloud before revealing the answer.

Reveal answer

Previous | Next question

---

## Part D — Mock practical examination

Try these without looking at the earlier formulas. These exercises cover the skills most directly related to your ten suggested assignments.

### Task 1 — Conditional formatting 1/9

Highlight sales in B2:B20 that exceed the average of B2:B20.

Attempt the task on a blank worksheet before revealing the solution.

Reveal solution

Previous | Next task

## Part E — Common mistakes that can cost marks

1. **Incorrect range references:** Writing `=B2>AVERAGE(B2:B20)` in conditional formatting can cause the average range to shift as the rule is evaluated. Lock the average range with `$`.
2. **Wrong row reference:** If a conditional-formatting rule starts at row 2, use row 2 in the formula. The formula's relative row must align with the first row of the selected range.
3. **Incorrect percentage values:** Excel stores 95% as `0.95`. Make sure the attendance values and formula thresholds use the same scale.
4. **Wrong VLOOKUP column index:** The index is counted from the first column of the selected lookup table, not from column A of the worksheet.
5. **Missing exact-match argument:** Use `FALSE` in VLOOKUP when an exact match is required.
6. **Nested IF in the wrong order:** Check higher thresholds before lower thresholds.

7. **Incorrect PivotTable aggregation:** Check whether a field is summarized by Sum, Count, or Average as required.
8. **Invalid named ranges:** In a dependent dropdown, the category names must match the defined names or be transformed appropriately.
9. **Confusing dates with text:** A date-looking entry may be stored as text. Verify the cell's actual value if date comparisons do not work.
10. **Forgetting to refresh:** Refresh a PivotTable after changing its source data.

## Part F — Your revision plan for today

Assuming you have around four hours available, use this schedule.

| | |
|---|---|
| 0:00–0:45 | **Conditional formatting and data validation**<br>Practise assignments 1–4; recreate two formula-based rules. |
| 0:45–1:30 | **Logical formulas and grades**<br>Practise IF, AND, OR, nested IF and attendance classification. |
| 1:30–2:10 | **Text functions and lookup functions**<br>Recreate email extraction, name formatting and IFERROR/IFNA. |
| 2:10–2:40 | **Statistical functions**<br>Practise AVERAGEIF, COUNTIF and SUMIF with a small dataset. |
| 2:40–3:20 | **PivotTables and charts**<br>Build a PivotTable from raw data, apply a Top 3 filter and insert a chart. |
| 3:20–4:00 | **Viva and weak areas**<br>Answer the flashcards aloud and repeat the mock tasks you missed. |

Before sleeping, make sure you can do these five things without notes:

- Create a conditional-formatting rule using a formula.
- Write a nested IF formula and explain its condition order.
- Use a text function to extract a substring.
- Use IFERROR or IFNA to handle a missing lookup.
- Create a PivotTable and filter the top three results.

One final piece of advice for the oral exam: **explain the logic, not just the syntax.** For example, when asked why you used AND in the overdue-order formula, say that the order must simultaneously be Pending and have a due date earlier than today. That demonstrates understanding rather than memorization.

Good luck with your practical and viva tomorrow. Focus on recreating the tasks yourself at least once; that is the most valuable remaining preparation.

YOU

explain me all the related concepts in all assignments with examples and how to recreate them in detail

# Complete Excel Practical & Viva Preparation — All Assignments

Your exam is tomorrow, so we'll prepare you to do three things for **every assignment**:

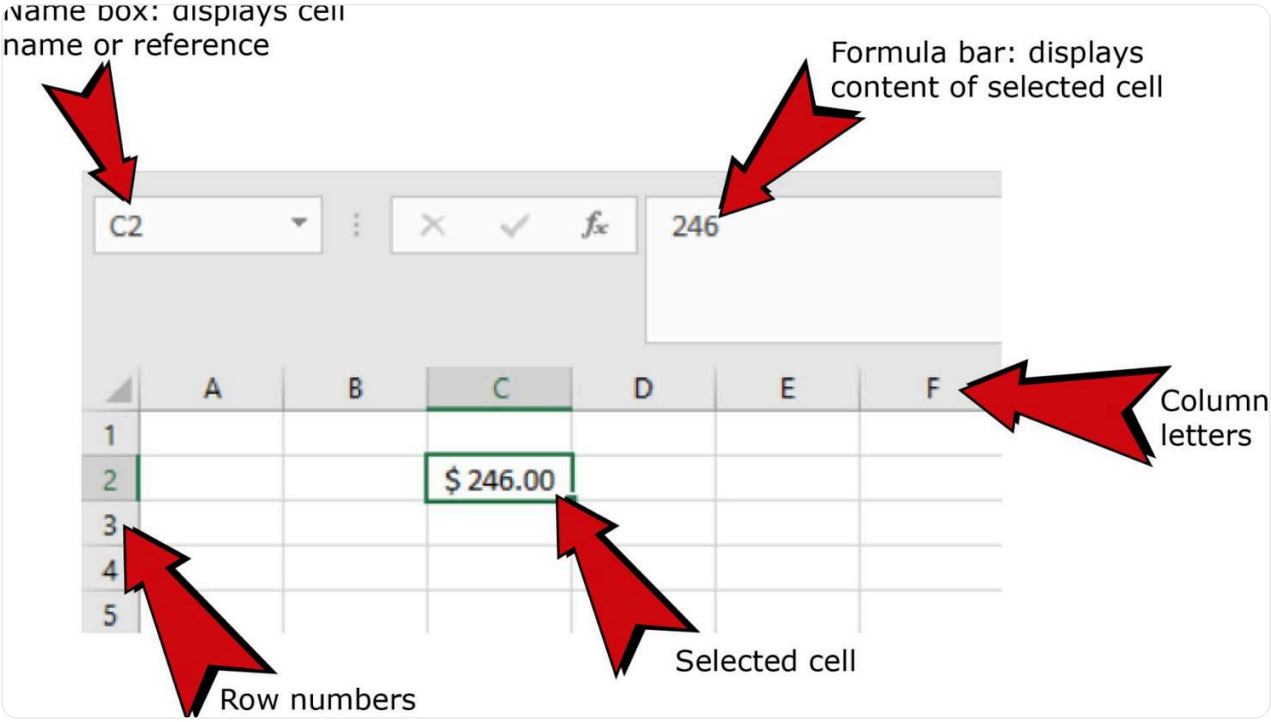
1. **Understand the concept** — what the Excel feature or function does and why it is used.
2. **Recreate the assignment** — which data to enter, which cells to select, which menu options to use, and which formulas to write.
3. **Answer the viva** — explain how the formula works, what its arguments mean, and what happens when the data changes.

I'll cover both your ten suggested assignments and the related topics in your four solved assignment workbooks: Excel essentials, advanced functions, PivotTables/charts/macros, and financial dashboards.

## First: the Excel fundamentals behind almost everything

Before working through the assignments, understand these concepts. They will make the practical tasks much easier.

## 1. Workbook, worksheet, cells and ranges




The Excel worksheet is a grid of rows and columns.

**Workbook**
The complete Excel file, such as `Sales.xlsx`.

**Worksheet**
An individual sheet inside the file.

**Cell**
One location, such as `B3`.

**Range**
A group of cells, such as `B2:B10`.

## 2. Formulas and functions

A **formula** is an expression that calculates a result. A **function** is a predefined operation that can be used inside a formula.

Examples:

```excel
=A2+B2
=SUM(A2:A10)
=AVERAGE(B2:B10)
```

The first adds two cells directly. The second and third use built-in functions.

Important rules:

- Begin formulas with `=`.

- Put text criteria in double quotes, such as `"Pending"`.
- Use `:` to specify a range.
- Use `&` to join text.
- Use parentheses to control which operations are evaluated together.

### 3. Relative, absolute and mixed cell references

This is one of the most important concepts for conditional formatting and copied formulas.

| Reference | Behavior when copied |
|---|---|
| `A2` | Column and row can change |
| `$A$2` | Neither column nor row changes |
| `$A2` | Column A stays fixed; row can change |
| `A$2` | Row 2 stays fixed; column can change |

Example: Suppose `B2` contains sales and `C2` contains a target.

```excel
=B2/C2
```

If you copy this formula down one row, it becomes `=B3/C3`.

But in this formula:

```excel
=B2/AVERAGE($B$2:$B$10)
```

the sales cell changes by row while the average range remains fixed.

**Viva question: Why do we use `$`?**

To lock a row or column reference when copying a formula or applying a formula-based conditional-formatting rule across multiple cells.

### 4. How to enter a formula correctly

1. Select the cell where you want the result.
2. Type `=`.
3. Enter the function or expression.
4. Press Enter.
5. If the formula must apply to multiple records, drag the fill handle down or double-click it when the adjacent data is continuous.

If you see `#NAME?`, Excel may not recognize a function or name. If you see `#VALUE!`, an argument may have an incompatible type. If you see `#N/A`, a lookup may not have found a matching value.

---

# Module 1 — Assignment 1: Excel essentials and conditional formatting

Your first workbook, `AE_A1(1).xlsx`, contains practical examples of sales monitoring, overdue orders, leave validation, dependent dropdowns, data cleaning, hospital records, and branch deposit comparison.

## Assignment 1 — Sales exception monitoring

### Concepts you must understand

- **AVERAGE:** calculates the arithmetic mean.
- **Conditional Formatting:** changes the appearance of cells based on a condition.
- **IF:** returns one result when a condition is true and another when false.
- **Achievement percentage:** compares actual sales with the target.
- **Exception monitoring:** highlights unusual results requiring management attention.

Suppose your table has the following columns:

| Column | Contents |
|---|---|
| A | Branch ID |
| B | Branch Location |
| C | Monthly Sales |
| D | Monthly Target |
| E | 60% of Target |
| F | Achievement % |
| G | Above Average? |
| H | Status Flag |

### How to recreate it

**Step 1: Enter the data.**

Enter a few branch IDs, city names, sales figures, and targets. Use at least five rows to test the formulas.

**Step 2: Calculate 60% of the target.**

In `E8`:

```excel
=D8*60%
```

This calculates the minimum acceptable sales threshold for that branch. Copy it down.

**Step 3: Calculate achievement percentage.**

In `F8`:

```excel
=C8/D8
```

Format column F as Percentage. If sales are ₹450,000 and the target is ₹400,000, achievement is 112.5%.

**Step 4: Classify sales relative to the average.**

In `G8`:

```excel
=IF(C8>AVERAGE($C$8:$C$19),"Yes","No")
```

This returns Yes when the branch's sales exceed the dataset average.

**Step 5: Classify branch status.**

In `H8`:

```excel
=IF(C8<E8,"CRITICAL SHORTFALL",IF(C8>=D8,"Target Met","Monitoring"))
```

The formula checks for a critical shortfall first. If sales are not below 60% of target, it checks whether the target has been met.

**Step 6: Apply conditional formatting.**

To highlight above-average sales:

1. Select `C8:C19`.
2. Go to Home → Conditional Formatting → New Rule.
3. Choose the formula-based rule.

4. Enter:

```excel
=C8>AVERAGE($C$8:$C$19)
```

5. Select a green fill and confirm.

To highlight sales below 60% of target, select the sales cells and use:

```excel
=C8<E8
```

Select a red fill.

**Important distinction:** The IF formulas produce text labels in cells; conditional formatting applies visual styles. They can be used together, but they are different features.

### Likely viva questions

**Why is the average range absolute?** Because the comparison must use the same dataset average for every branch.

**What happens when a new branch is added?** The formulas and formatting range may need to be extended. An Excel Table can make expansion easier.

**What does achievement above 100% mean?** Actual sales exceed the target.

---

## Assignment 2 — Overdue order tracking dashboard

### Concepts

- Date comparisons
- `TODAY()`
- `AND()`
- Conditional formatting across entire rows
- `IF()` for status labels
- Absolute references for a reference date

Use this structure:

| A | B | C | D | E | F | G |
|---|---|---|---|---|---|---|
| Order ID | Customer | Order Date | Due Date | Order Status | Delivery Date | Delivery Status |

Assume the reference date is stored in `I6`. You can enter a fixed date for practice or use `=TODAY()` for a dynamic date.

**Step 1: Calculate days overdue.**

If the order is pending and the due date has passed, calculate the number of overdue days. In `H8`:

```excel
=IF(E8="Completed",F8-D8,IF($I$6>D8,$I$6-D8,0))
```

This example uses the due date in D as the comparison date for the calculation. If your intended calculation is specifically days overdue beyond the due date, use:

```excel
=IF(E8="Completed",MAX(0,F8-D8),IF(E8="Pending",MAX(0,$I$6-D8),0))
```

The second version follows the sample workbook's date structure only when its intended interpretation is days since the due date. In a real overdue tracker, the due date should be the reference for overdue days, not the order date.

A cleaner formula for **days overdue beyond the due date** is:

```excel
=IF(E8="Completed",MAX(0,F8-D8),IF(E8="Pending",MAX(0,$I$6-D8),0))
```

Here, `MAX(0,...)` prevents negative values.

**Step 2: Create a priority-action label.**

```excel
=IF(AND(E8="Pending",D8<$I$6),"IMMEDIATE ESCALATION",IF(E8="Completed","Closed","On Schedule"))
```

**Step 3: Highlight overdue rows.**

Select `A8:I19`, then create a formula-based conditional-formatting rule:

```excel
=AND($E8="Pending",$D8<$I$6)
```

Choose red fill. The `$E` and `$D` lock the columns while the row number changes.

**Step 4: Add separate status formatting.**

Create rules for Pending, Completed, and Delayed if those values exist in your status column. For example:

```excel
=$E8="Pending"
```

Use yellow for Pending, green for Completed, and red for Delayed.

### Likely viva questions

- Why use AND? Both conditions must be true.
- Why use a reference-date cell instead of typing a date into every formula? It centralizes the date, making testing and updates easier.
- Why use `<` rather than `<=`? To distinguish dates strictly before today from dates due today.
- What is the difference between order date and due date? Order date records when the order was placed; due date indicates when it should be completed.

---

## Assignment 3 — Employee leave application validation

### Concepts

- Date validation
- List validation
- Formula-based validation
- Date arithmetic
- IF-based error/status labels

Suppose the columns are:

- D: Start Date
- E: End Date
- F: Leave Type
- G: Requested Days
- H: Available Balance
- I: Validation Status

**Step 1: Calculate leave duration.**

In `G8`:

```excel
=E8-D8+1
```

The `+1` counts both the start and end dates. For example, 10 May through 12 May is three calendar days.

If weekends and holidays should be excluded, use `NETWORKDAYS` with an appropriate holiday list instead.

**Step 2: Validate dates.**

Select the start-date cells and choose Data → Data Validation → Allow: Date → Between.

For an academic or financial year of 1 April 2026 to 31 March 2027, use those two dates as the boundaries. Repeat for the end-date column.

**Step 3: Create a Leave Type dropdown.**

1. Enter `Casual Leave`, `Sick Leave`, and `Earned Leave` in a small source list.
2. Select the Leave Type cells.
3. Choose Data → Data Validation → Allow: List.
4. Select the source cells.

**Step 4: Check the leave balance.**

In `I8`:

```excel
=IF(G8>H8,"EXCEEDED BALANCE","VALID - APPROVED")
```

This is a classification formula. It does not itself block invalid leave requests.

To actually restrict entries based on the balance, use a suitable Custom validation rule. For example, if the requested duration is in G and the available balance in H:

```excel
=G8<=H8
```

Configure an error alert to prevent the user from accepting an invalid request.

**Viva:** Data validation controls what users may enter; formulas calculate results and flag conditions. They can complement each other.

---

## Assignment 4 — Dependent category/subcategory dropdowns

### Concepts

- Data Validation
- Named ranges

- `INDIRECT()`
- Lookup-style references
- Source lists

Suppose your source lists contain:

| Electronics | Stationery | Furniture |
|---|---|---|
| Laptops | Notebooks | Chairs |
| Smartphones | Pens | Tables |
| Monitors | Files | Desks |

### Recreate it step by step

**Step 1:** Put the three category names in a separate source area.

**Step 2:** Select each subcategory list and define its name using the Name Box. For example, name the electronics subcategory range `Electronics`.

**Step 3:** Create a Category dropdown in `B8`.

- Select `B8`.
- Data → Data Validation → List.
- Set Source to the category list.

**Step 4:** Create a Subcategory dropdown in `C8`.

- Select `C8`.
- Data → Data Validation → List.
- Set Source to:

```excel
=INDIRECT(B8)
```

When B8 contains `Electronics`, Excel resolves the named range called Electronics.

**Step 5:** Test all categories.

Select each category and verify that only its subcategories appear.

**Step 6:** Calculate the line value.

If quantity is in E and unit price is in F:

```
=E8*F8
```
EXCEL

### Viva questions

- What is a named range? A meaningful name assigned to a cell or range.
- What does INDIRECT do? Converts a text reference into a reference Excel can use.
- Why use dependent dropdowns? To reduce invalid category-subcategory combinations and speed up data entry.
- What is the limitation? The category value must resolve to a valid named range. Names containing spaces require special handling.

---

## Assignment 5 — Student admission data cleaning

### Concepts

- Sorting
- Multi-level sorting
- Filtering
- `RANK()`
- `TRIM()`
- Threshold classification

**Step 1: Sort by multiple fields.**

Select the dataset, then go to Data → Sort.

Add levels in this order:

1. Program — A to Z
2. City — A to Z
3. Admission Score — Largest to Smallest

This groups the records by program, then city, and orders candidates by score within those groups.

**Step 2: Filter the data.**

Select the headers and choose Data → Filter. Use the dropdowns to select the required cities and filter the score column to values greater than or equal to the specified threshold.

**Step 3: Rank students.**

If scores are in `E8:E22`, enter in F8:

```excel
=RANK(E8,$E$8:$E$22,0)
```

The final argument `0` requests descending rank: the largest score receives rank 1. In current Excel, `RANK.EQ` is also available.

**Step 4: Classify applicants.**

```excel
=IF(E8>=80,"SHORTLISTED","Below Threshold")
```

**Step 5: Clean text fields.**

```excel
=TRIM(B8)
```

This removes leading and trailing spaces and reduces repeated internal spaces to a single space.

### Viva questions

- What is the difference between sorting and filtering?
- Why is `$E$8:$E$22` locked in the ranking formula?
- What happens when two students have the same score? They receive the same rank with `RANK.EQ`, and subsequent ranks may be skipped.
- Why use TRIM? To remove unwanted spaces that can cause matching and data-quality problems.

---

## Assignment 6 — Hospital records and large datasets

### Concepts

- Freeze Panes
- Remove Duplicates
- AutoFilter
- Formula-based action labels

- Data integrity

**Step 1: Freeze headers and key columns.**

Suppose the headers are in row 7 and columns A and B contain Patient ID and Patient Name.

To freeze the first two columns and the top seven rows:

1. Select cell `C8`.
2. Go to View → Freeze Panes → Freeze Panes.

Excel freezes all rows above the selected cell and all columns to its left. This keeps rows 1–7 and columns A–B visible while scrolling.

**Step 2: Remove duplicates.**

1. Make a backup of the data.
2. Select the dataset.
3. Go to Data → Remove Duplicates.
4. Select Patient ID and Visit Date if duplicates mean repeated patient visits on the same date.
5. Confirm the result.

Do not automatically remove every repeated Patient ID if the same patient can legitimately have multiple visits.

**Step 3: Filter unpaid bills.**

Use Data → Filter, then select `Unpaid` in the Payment Status column.

**Step 4: Create an action label.**

If payment status is in H8:

```excel
=IF(H8="Unpaid","FOLLOW-UP RECOVERY","Clear")
```

### Viva questions

- Why use Freeze Panes? To retain context when working with large datasets.
- What is a duplicate record? A record that repeats according to the fields used to identify duplicates.
- Does filtering delete records? No.

- Why is Patient ID alone sometimes insufficient for duplicate detection? A patient may have multiple legitimate visits.

---

## Assignment 7 — AI-enhanced data review in Excel

This is a conceptual case study in your syllabus rather than just a formula exercise.

### Concepts

- Data types
- Automated suggestions
- Data enrichment
- Pattern detection
- Human verification

**How to recreate a simple version**

1. Create a sales dataset with city, product, quantity, and revenue.
2. Select the city column.
3. If supported in your Excel version, use Data → Data Types → Geography.
4. Examine the linked information offered by Excel.
5. Create a PivotTable summarizing sales by city.
6. Identify high- and low-performing cities and investigate unusual values.

Some Excel AI features require a compatible Microsoft 365 version, account, and internet connection. If unavailable, explain the intended workflow and demonstrate the analysis with ordinary Excel tools.

**Viva:** AI-enabled features can accelerate analysis or enrich records, but the results must be verified. AI suggestions do not guarantee that every inferred value is correct.

---

## Assignment 8 — Branch deposit performance comparison

This task uses percentage calculations, nested IF, and conditional formatting.

Suppose:

- D8 = Deposit Target
- E8 = Actual Deposit
- F8 = Achievement Percentage
- G8 = Performance Tier
- H8 = Managerial Action

**Step 1: Calculate achievement percentage.**

```excel
=E8/D8
```

Format as Percentage.

**Step 2: Assign performance tiers.**

```excel
=IF(F8>110%,"Exceeds (>110%)",IF(F8>=90%,"Meets (90%-110%)","Below (<90%)"))
```

The first condition checks values above 110%. The second checks values from 90% through 110%, inclusive. Everything else falls below 90%.

**Step 3: Add an action formula.**

```excel
=IF(F8>110%,"Award Bonus",IF(F8>=90%,"Maintain Standard","Review Strategy"))
```

**Step 4: Apply conditional formatting.**

Select the achievement percentage cells and create three rules:

```excel
=F8>110%
```

```excel
=AND(F8>=90%,F8<=110%)
```

```excel
=F8<90%
```

Use green, yellow, and red respectively.

**Viva:** Why is the order of the nested IF conditions important? Because Excel returns the result of the first TRUE condition. Testing the wrong thresholds first can assign the wrong tier.

---

# Module 2 — Assignment 2: Advanced functions and data analysis

Your `AE_A2.xlsx` workbook covers employee appraisal, student results, ticket priority, sales incentives, product lookups, text standardization, invoice numbers, deadlines, salary analysis, and other formula-based business cases.

## Assignment 9 — Employee appraisal using IF, AND and OR

This assignment is about turning several business rules into one classification formula.

Suppose:

| Column | Field |
|---|---|
| C | Performance Score |
| D | Attendance |
| E | Project Status |
| F | Appraisal Classification |

The workbook uses these rules:

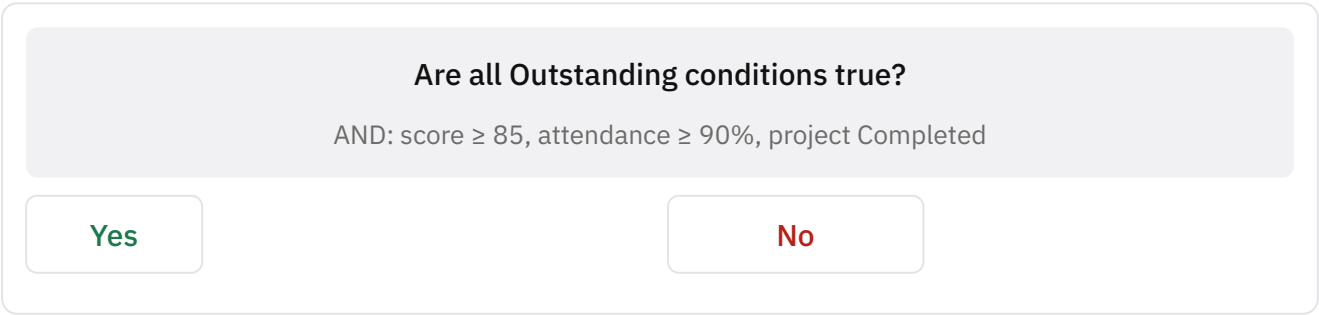
- Outstanding: score at least 85, attendance at least 90%, and projects Completed.
- Needs Improvement: score below 60, or attendance below 75%, or project status Overdue.
- Satisfactory: all other cases.

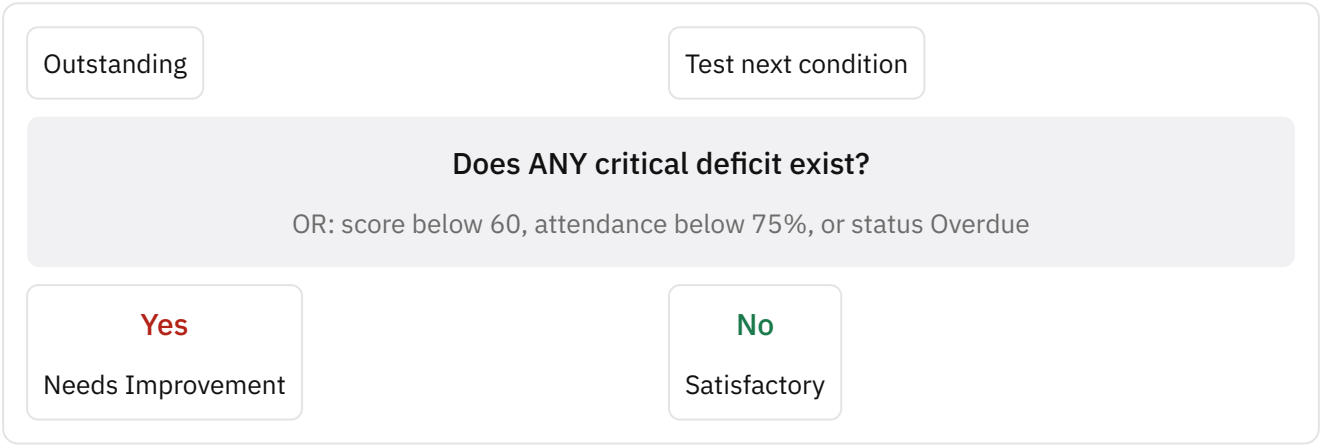
Enter this in `F8`:

```excel
=IF(AND(C8>=85,D8>=90%,E8="Completed"),
"Outstanding",
IF(OR(C8<60,D8<75%,E8="Overdue"),
"Needs Improvement",
"Satisfactory"))
```

If necessary, paste it as one continuous line.

### How the formula is evaluated






The workbook also calculates bonus eligibility in G8:

```excel
=IF(F8="Outstanding","Yes - 20%",IF(F8="Satisfactory","Yes - 10%","No Bonus"))
```

And managerial action in H8:

```excel
=IF(F8="Outstanding","High potential talent",IF(F8="Needs Improvement","Schedule PIP review","Stable contributor"))
```

Copy all formulas down.

**Viva:** AND requires all conditions to be true. OR requires at least one. Nested IF lets you test several categories in sequence.

## Assignment 10 — Student result processing with nested IF

This is an excellent exercise for combining arithmetic, percentages, minimum marks, and nested conditions.

Suppose five subjects are in C8:G8, each out of 100.

### Step 1: Calculate total marks

In H8:

```excel
=SUM(C8:G8)
```

Maximum possible marks are 500.

### Step 2: Calculate percentage

In I8:

```excel
=H8/500
```

Format the cell as Percentage.

### Step 3: Assign a grade

Your workbook uses the following percentage thresholds:

| Percentage | Grade |
|---|---|
| 90% or more | O |
| 80%–below 90% | A+ |
| 70%–below 80% | A |
| 60%–below 70% | B+ |
| 55%–below 60% | B |
| 50%–below 55% | C |
| Below 50% | P |

In J8:

```excel
=IF(I8>=90%,"O",
IF(I8>=80%,"A+",
IF(I8>=70%,"A",
IF(I8>=60%,"B+",
IF(I8>=55%,"B",
IF(I8>=50%,"C","P"))))))
```

The thresholds are tested from highest to lowest.

### Step 4: Identify distinction

The workbook requires at least 75% overall and at least 50 marks in every subject.

In K8:

```excel
=IF(AND(I8>=75%,MIN(C8:G8)>=50),"DISTINCTION","Standard Pass")
```

`MIN(C8:G8)` finds the lowest subject mark.

### Step 5: Flag re-evaluation eligibility

In L8:

```excel
=IF(AND(I8>=45%,I8<50%),"Eligible for Re-evaluation","Not Applicable")
```

This is the workbook's stated illustrative rule. Actual university eligibility may differ.

### Viva questions

- Why calculate total before percentage? Percentage requires the total earned and the maximum possible marks.

- Why use MIN? To check whether any subject falls below the required minimum.
- Why use AND for distinction? Both the overall percentage and the minimum subject-mark conditions must be satisfied.
- What is the difference between marks and percentage? Marks are the raw score; percentage expresses the score relative to the maximum possible score.

---

## Assignment 11 — Customer support ticket priority analyzer

This assignment uses logical functions to categorize support tickets.

Suppose:

- C8 = Customer Type
- D8 = Severity
- E8 = Pending Days
- F8 = Resolution Status
- G8 = Priority Label

The workbook's rules are:

- Resolved tickets → Low Priority.
- Otherwise, Critical severity, more than three pending days, or Enterprise customer → High Priority.
- Otherwise, Major severity or at least two pending days → Medium Priority.
- Otherwise → Low Priority.

Formula in G8:

```excel
=IF(F8="Resolved","Low Priority",
IF(OR(D8="Critical",E8>3,C8="Enterprise"),"High Priority",
IF(OR(D8="Major",E8>=2),"Medium Priority","Low Priority")))
```

In H8, the workbook checks SLA breach risk:

```excel
=IF(AND(G8="High Priority",E8>=3),"SLA BREACH RISK","SLA Compliant")
```

In I8, it assigns an action:

```excel
=IF(G8="High Priority","Escalate to Tier 2",IF(G8="Medium Priority","Standard Queue","Close Ticket"))
```

**Important logic detail:** Because the formula checks Resolved first, a resolved Enterprise ticket is Low Priority despite being an Enterprise customer.

**Viva:** What is an SLA? A Service Level Agreement, which defines expected service standards such as response or resolution times. Priority formulas help identify tickets that may need escalation.

---

## Assignment 12 — Sales incentive calculator using VLOOKUP

This assignment introduces lookup tables, approximate matching, and commission calculations.

Suppose the reference table is in K8:L12.

| Sales slab (₹) | Incentive rate |
|---:|---:|
| 0 | 0% |
| 1,000,000 | 5% |
| 2,000,000 | 8% |
| 3,500,000 | 12% |
| 5,000,000 | 15% |

Your sales value is in C8.

### Step 1: Retrieve the slab rate

In D8:

```excel
=VLOOKUP(C8,$K$8:$L$12,2,TRUE)
```

The arguments are:

- `C8`: the value to search for.
- `$K$8:$L$12`: the reference table.
- `2`: return the value from the second column.
- `TRUE`: approximate match.

For approximate matching, the first column of the lookup table must be sorted in ascending order. Excel finds the largest threshold that is less than or equal to the sales amount.

For example, sales of ₹2,800,000 match the ₹2,000,000 slab and return an 8% rate.

### Step 2: Calculate the incentive

In E8:

```excel
=C8*D8
```

For ₹2,800,000 at 8%, the incentive is ₹224,000.

### Step 3: Classify performance

```excel
=IF(D8>=12%,"Top Performer",IF(D8>=8%,"Core Contributor","Developing"))
```

### Step 4: Determine payout status

```excel
=IF(E8>0,"Approved for Payroll","Base Salary Only")
```

**Viva:** Why use a lookup table instead of nested IF for all sales slabs? The rates can be updated in one place, reducing repeated formulas and maintenance effort.

---

## Assignment 13 — Product price retrieval using INDEX and MATCH

This task teaches two functions working together.

Suppose:

- A14:A23 contains product codes.
- B14:B23 contains product names.
- D14:D23 contains suppliers.
- E14:E23 contains prices.
- C8 is the input cell for the product code.

### Step 1: Retrieve product name

In C9:

```excel
=INDEX(B14:B23,MATCH(C8,A14:A23,0))
```

### Step 2: Retrieve supplier

In C10:

```excel
=INDEX(D14:D23,MATCH(C8,A14:A23,0))
```

### Step 3: Retrieve price

In C11:

```excel
=INDEX(E14:E23,MATCH(C8,A14:A23,0))
```

### How INDEX and MATCH work

- `MATCH(C8,A14:A23,0)` finds the exact position of the product code.
- `INDEX(B14:B23,position)` returns the value at that position in the product-name range.

Suppose the product code is found in the third position of the lookup range. MATCH returns 3, and INDEX returns the third product name.

To handle missing product codes, wrap the formula in IFNA:

```excel
=IFNA(INDEX(B14:B23,MATCH(C8,A14:A23,0)),"Product Not Found")
```

**Viva:** Why is INDEX-MATCH more flexible than VLOOKUP? It can return values from columns to the left or right of the lookup column, and its independent ranges can be less sensitive to inserted columns than a hard-coded VLOOKUP column index.

---

## Assignment 14 — Library email and text analysis

This assignment uses `FIND`, `MID`, `LEN`, `RIGHT`, `LEFT`, `UPPER`, and concatenation.

Suppose:

- A8 = Member ID
- B8 = Raw Student Name
- C8 = Email Address

### Extract the email domain

In D8:

```excel
=MID(C8,FIND("@",C8)+1,LEN(C8))
```

For `aarav.sharma@pccoepune.org`, the result is `pccoepune.org`.

`FIND` locates @, `+1` moves to the first character after it, and `MID` extracts the remaining text.

### Standardize names

In E8:

```excel
=UPPER(RIGHT(B8,LEN(B8)-FIND(" ",B8)))&", "&LEFT(B8,FIND(" ",B8)-1)
```

For `Aarav Sharma`, the result is `SHARMA, Aarav`.

### Count member ID characters

In F8:

```excel
=LEN(A8)
```

### Validate the ID length

In G8:

```excel
=IF(F8=8,"Valid ID","Invalid Length")
```

This example expects eight characters, such as `LIB-1001`.

**Viva:** `LEN` counts spaces and punctuation too. `FIND` is case-sensitive, while `SEARCH` is not. Both can return an error if the search text is missing.

---

## Assignment 15 — Invoice number standardization

The source workbook contains invoice formats such as `INV-2026-PUN-045` and `BILL/26/MUM/102`.

This assignment combines `SUBSTITUTE`, `LEFT`, `MID`, `RIGHT`, and concatenation.

### Step 1: Standardize delimiters

In C8:

```excel
=SUBSTITUTE(B8,"/","-")
```

This replaces every `/` with `-`.

### Step 2: Extract the city code

The exact character position depends on the invoice pattern. For the example `INV-2026-PUN-045`:

```excel
=MID(C8,10,3)
```

This returns `PUN`.

For a pattern like `BILL-26-MUM-102`, the city code occurs at a different position. The workbook uses conditional logic to handle the different prefixes.

### Step 3: Extract the year

For `INV-2026-PUN-045`:

```excel
=MID(C8,5,4)
```

For a two-digit year such as `26`, one possible conversion is:

```excel
="20"&MID(C8,6,2)
```

Use the formula matching the actual input structure.

### Step 4: Extract the serial number

```excel
=RIGHT(C8,3)
```

### Step 5: Rebuild the standard format

```excel
="INV-"&E8&"-"&D8&"-"&F8
```

This creates `INV-2026-PUN-045`, assuming E8 is the year, D8 the city code, and F8 the serial number.

**Viva:** Why use SUBSTITUTE before extracting? It standardizes the separators. Why can MID formulas fail? Because fixed character positions depend on the input format; inconsistent lengths require more robust parsing logic.

---

## Assignment 16 — Project deadline monitoring

This assignment introduces date arithmetic and four useful date functions.

Suppose:

- D8 = Start Date
- E8 = Deadline Date
- I6 = Reference Date

### Project age

```excel
=DATEDIF(D8,$I$6,"d")
```

Returns the number of complete days between the start date and the reference date.

To calculate years of service instead, the unit would be `"y"`.

### Remaining days

```excel
=E8-$I$6
```

Positive means days remaining; zero means due today; negative means overdue.

### Month-end review date

```excel
=EOMONTH(E8,0)
```

Returns the last day of the month containing the deadline.

For the end of the following month, use `=EOMONTH(E8,1)`.

### Deadline warning within seven days

If the reference date is in I6 and the deadline is in E8:

```excel
=AND(E8>=$I$6,E8<=$I$6+7)
```

This flags deadlines from today through seven days ahead, inclusive.

**Viva:** Excel stores dates as serial numbers, which is why date subtraction works. DATEDIF is useful for elapsed time, and EOMONTH is useful for month-end reporting.

---

## Assignment 17 — Employee service tenure

Suppose A8 contains the joining date and B8 the current/reference date.

Calculate complete years of service:

```excel
=DATEDIF(A8,B8,"y")
```

Calculate complete months remaining after full years:

```excel
=DATEDIF(A8,B8,"ym")
```

Display both:

```excel
=DATEDIF(A8,B8,"y")&" years, "&DATEDIF(A8,B8,"ym")&" months"
```

To identify employees reaching five years of service this year, the calculation needs to consider both the joining date and the current date. A simple check for exactly five complete years is:

```excel
=DATEDIF(A8,TODAY(),"y")=5
```

But that identifies employees who currently have five complete years, not necessarily employees whose fifth anniversary occurs during the calendar year. For the latter, compare the joining-date anniversary with the start and end of the year.

**Viva:** What is the difference between `"y"` and `"ym"`? `"y"` returns complete years; `"ym"` returns the remaining months after complete years.

---

## Assignment 18 — Error-proof student lookup

This task is similar to product lookup but focuses on missing values.

Suppose A2:A100 contains roll numbers and B2:B100 contains names. E2 is the roll-number input.

Using XLOOKUP:

```excel
=XLOOKUP(E2,A2:A100,B2:B100,"Student Not Found")
```

Using VLOOKUP:

```excel
=IFNA(VLOOKUP(E2,A2:B100,2,FALSE),"Student Not Found")
```

Using INDEX-MATCH:

```excel
=IFNA(INDEX(B2:B100,MATCH(E2,A2:A100,0)),"Student Not Found")
```

**Viva:** Why handle lookup errors? A meaningful message is easier for users to understand than a raw `#N/A` error. IFNA is appropriate when you want to handle missing matches while allowing other formula errors to remain visible.

---

## Assignment 19 — Salary analysis

Suppose:

- B8:B100 contains Department.
- C8:C100 contains Salary.
- D8:D100 contains Designation.

### Average salary by department

```excel
=AVERAGEIF(B8:B100,"IT",C8:C100)
```

### Count employees above a threshold

If salaries are in C8:C100:

```excel
=COUNTIF(C8:C100,">50000")
```

If the threshold is stored in F2:

```excel
=COUNTIF(C8:C100,">"&F2)
```

The `&` combines the comparison operator with the value in F2.

### Total salary for a designation

```excel
=SUMIF(D8:D100,"Manager",C8:C100)
```

**Viva:** `AVERAGEIF` calculates a conditional average, `COUNTIF` counts matching cells, and `SUMIF` totals matching values. These are useful for departmental analysis without creating a PivotTable.

---

## Assignment 20 — Attendance risk analysis

Suppose student attendance percentages are in B2:B51.

### Average attendance

```excel
=AVERAGE(B2:B51)
```

### Count students below 75%

```excel
=COUNTIF(B2:B51,"<75%")
```

### Classify attendance

For an illustrative rule of Safe at 85% or more, Warning at 75% to below 85%, and Critical below 75%:

```excel
=IF(B2>=85%,"Safe",IF(B2>=75%,"Warning","Critical"))
```

Use the exact thresholds provided in the exam question.

**Viva:** Why use COUNTIF? It counts the students satisfying the threshold without manually checking each record. Why use nested IF? Because the classification has multiple possible categories.

---

## Assignment 21 — Expense review with error handling

Suppose:

- A2:A50 = Expense Category
- B2:B50 = Actual Amount
- C2:C50 = Budget

### Calculate variance

```excel
=B2-C2
```

Positive means overspending for an expense.

### Categorize budget status

```excel
=IF(B2>C2,"Over Budget","Within Budget")
```

### Sum expenses by category

```excel
=SUMIF(A2:A50,"Travel",B2:B50)
```

### Handle invalid division

```excel
=IFERROR(B2/C2,"Check Budget")
```

This prevents a raw error from being displayed if C2 is zero or the calculation otherwise fails.

**Viva:** What is the risk of using IFERROR everywhere? It may hide genuine formula mistakes. Use it when you have a sensible fallback for the errors that could occur.

## Assignment 22 — Regional sales comparison

Suppose the sales table contains Region and Actual Sales, while a separate target table contains Region and Target.

### Retrieve a target

With XLOOKUP:

```excel
=XLOOKUP(A2,$H$2:$H$10,$I$2:$I$10,"Target Missing")
```

Here A2 contains the region, H contains region names, and I contains targets.

### Compare actual sales with target

```excel
=IF(B2>=C2,"Achieved",IF(B2>=90%*C2,"Near Target","Below Target"))
```

This classifies a region as Achieved if it meets or exceeds the target, Near Target if it reaches at least 90% of target, and Below Target otherwise.

### Calculate regional statistics

Average sales for a region:

```excel
=AVERAGEIF(A2:A100,"Western",B2:B100)
```

Maximum sales for a region:

```excel
=MAXIFS(B2:B100,A2:A100,"Western")
```

`MAXIFS` is available in modern Excel versions. If unavailable, use a PivotTable or another compatible approach.

**Viva:** Why use a separate target table? It keeps target values centralized and allows the same lookup formula to work across many regions.

---

# Module 3 — Assignment 3: PivotTables, charts, slicers and macros

This module is less about long formulas and more about arranging, summarizing, and presenting data. You should practise the menu paths as well as understand the purpose of each tool.

## Assignment 23 — Regional sales performance using a PivotTable

Suppose your source data contains:

| Date | Region | Product Category | Salesperson | Quantity | Sales Amount |
|---|---|---|---|---|---|
| 1 Oct | West | Electronics | Ravi | 5 | 250000 |
| 2 Oct | North | Furniture | Neha | 3 | 90000 |
| 3 Oct | West | Furniture | Ravi | 2 | 40000 |
| 4 Oct | South | Electronics | Amit | 4 | 200000 |

### How to recreate it

#### Step 1 — Prepare the source data

- Ensure every column has a unique heading.
- Remove completely blank rows within the dataset.
- Check that Sales Amount is numeric.
- Select a cell inside the data.

Optionally press **Ctrl+T** to convert the range to an Excel Table.

#### Step 2 — Insert a PivotTable

1. Choose Insert → PivotTable.
2. Check the selected table or range.
3. Select New Worksheet.
4. Click OK.

#### Step 3 — Arrange the fields

In the PivotTable Fields pane:

- Drag Region to Rows.
- Drag Sales Amount to Values.
- Drag Product Category to Columns if you want category-wise comparisons.
- Drag Date or Salesperson to Filters if you want to filter by a particular date or salesperson.

#### Step 4 — Check the calculation

The Values area should show **Sum of Sales Amount**.

If it shows Count, open the field's dropdown, choose Value Field Settings, and select Sum. If Sum is unavailable or incorrect, inspect the source data for numbers stored as text.

**Step 5 — Show the Top 3 regions**

1. Open the dropdown for the Region row labels.
2. Select Value Filters → Top 10.
3. Change 10 to 3.
4. Select Top, Items, and the Sum of Sales Amount field.
5. Click OK.

**Step 6 — Format the report**

Give the PivotTable a meaningful title, format totals as currency, and sort by total sales if appropriate.

### What a PivotTable is actually doing

A PivotTable groups repeated region names and aggregates their sales. Instead of manually calculating each regional total, Excel performs the grouping and summarization for you.

For example, two West transactions of ₹250,000 and ₹40,000 contribute ₹290,000 to the West total.

**Viva questions**

- What is the difference between Rows and Columns in a PivotTable? Both group data, but they organize the categories along different axes.
- What is the Values area used for? Aggregations such as Sum, Count, Average, Max, and Min.
- What is a Value Filter? A filter that uses aggregated results, such as showing only regions with the top three sales totals.
- Why refresh a PivotTable? To update its summary after the underlying data changes.

---

## Assignment 24 — Product category trend analysis using a PivotChart

The aim is to compare monthly sales for multiple product categories over six months.

### How to recreate it

**Step 1:** Create a transaction dataset containing Date, Product Category, and Sales Amount.

**Step 2:** Insert a PivotTable.

**Step 3:** Arrange the fields:

- Date → Rows
- Product Category → Columns
- Sales Amount → Values

**Step 4:** Group dates by month if the dates are individual transaction dates.

Depending on the Excel version, right-click a date in the PivotTable and choose Group, then select Months. If the dataset spans multiple years, include Years too so January from different years is not combined incorrectly.

**Step 5:** Insert a PivotChart.

Select the PivotTable, then choose PivotTable Analyze → PivotChart, or use Insert → PivotChart depending on your version.

Choose a line chart for comparing trends over time or a clustered column chart for comparing monthly category totals.

**Step 6:** Interpret the chart.

- An upward trend suggests sales are growing.
- A downward trend suggests declining sales.
- Sharp irregular movements may indicate seasonal demand, promotions, supply disruptions, or unusual transactions.

A chart shows patterns; it does not automatically prove their cause.

**Viva:** A PivotChart is linked to a PivotTable, so its displayed data responds to compatible PivotTable filtering and changes.

---

## Assignment 25 — Student placement analysis with grouping and filtering

The objective is to analyze placements by month and company, count students, and calculate average packages.

### How to recreate it

**Step 1:** Prepare fields such as Student ID, Company, Placement Date, and Package.

**Step 2:** Insert a PivotTable.

**Step 3:** Place fields as follows:

- Company → Rows
- Placement Date → Rows, below Company, if you want a company-by-date breakdown
- Student ID → Values, summarized by Count
- Package → Values, summarized by Average

Alternatively, place the date above the company field if you want to group by month first and companies within each month.

**Step 4:** Group placement dates by Months and Years.

**Step 5:** Filter packages above the selected threshold. For record-level filtering, apply a source-data filter before the PivotTable or use a suitable report filter. A Value Filter can instead filter aggregated PivotTable results.

**Viva:** What is the difference between Count of Student ID and Average of Package? The first measures placement records; the second calculates the average compensation represented by the selected records.

---

## Assignment 26 — Inventory movement dashboard using a PivotTable and slicers

Suppose the source data has:

- Item Name
- Category
- Supplier
- Inward Quantity
- Outward Quantity
- Month

### How to recreate it

**Step 1:** Insert a PivotTable.

**Step 2:** Arrange the fields:

- Category → Rows
- Month → Columns
- Inward Quantity → Values, summarized by Sum
- Outward Quantity → Values, summarized by Sum
- Supplier → Filters, if you want a basic supplier filter

**Step 3:** Calculate net movement if required.

In the source dataset, create a Net Movement column:

```excel
=D2-E2
```

This assumes D2 is Inward Quantity and E2 is Outward Quantity.

Add Net Movement to the PivotTable Values area.

**Step 4:** Add slicers.

1. Select the PivotTable.
2. Choose PivotTable Analyze → Insert Slicer.
3. Select Supplier and Category.
4. Click OK.
5. Move the slicers beside the PivotTable and test them.

Slicers provide clickable filter buttons, allowing users to interactively explore the dashboard.

**Viva:** Inward quantity represents incoming stock; outward quantity represents stock leaving inventory. Net movement is inward minus outward, but it is not necessarily the same as closing stock unless opening stock is also accounted for.

---

## Assignment 27 — Attendance monitoring report using a PivotTable and chart

Suppose the dataset contains Student, Class, Subject, Present Days, and Total Days.

### Step 1: Calculate attendance percentage

In the source data, if Present Days is E2 and Total Days is F2:

```excel
=E2/F2
```

Format the result as Percentage.

### Step 2: Create a PivotTable

For an average of recorded attendance percentages:

- Class → Rows
- Subject → Columns
- Attendance Percentage → Values, summarized by Average

If you have only Present Days and Total Days, summing those fields separately and dividing the totals may be more appropriate than averaging individual percentages, depending on the required definition of attendance.

### Step 3: Create a PivotChart

Select the PivotTable and insert a clustered column chart to compare attendance percentages across subjects.

### Step 4: Interpret the result

Identify subjects or classes with lower attendance and discuss where intervention may be needed.

**Viva:** Why is the choice of aggregation important? Because Average of percentages and the percentage calculated from total present days divided by total possible days can differ when students have different numbers of recorded days.

---

## Assignment 28 — Macro-based monthly sales report automation

A macro records or runs a series of steps that would otherwise be performed repeatedly.

### Concepts

- Macro recording
- VBA

- Relative versus absolute macro recording
- Developer tab
- Macro-enabled workbooks
- Automation

### How to recreate a simple macro

**Step 1:** Open a practice workbook containing monthly sales.

**Step 2:** Enable the Developer tab if it is not visible.

In many desktop Excel versions:

1. File → Options.
2. Customize Ribbon.
3. Tick Developer.
4. Click OK.

**Step 3:** Record the macro.

1. Go to Developer → Record Macro.
2. Name it `FormatMonthlyReport`.
3. Choose where to store it, usually This Workbook.
4. Click OK.

**Step 4:** Perform the actions to automate.

For example:

- Bold the header row.
- Apply a fill color.
- Format sales as currency.
- Add a total using SUM.
- Adjust column widths.

**Step 5:** Stop recording.

Choose Developer → Stop Recording.

**Step 6:** Run it again.

Make a fresh copy of the raw data, then run the macro and verify that the expected formatting is applied.

**Step 7:** Save correctly.

Use `.xlsm` if the workbook needs to retain VBA macros.

### Important limitations

A recorded macro may contain fixed cell references. If your report grows or changes layout, the macro may not behave as intended. More advanced automation may require editing VBA.

Never enable macros in an untrusted workbook merely to view its contents.

#### Viva questions

- What is VBA? Visual Basic for Applications, the programming language used for Office automation.
- What is the difference between a macro and a formula? A formula calculates a cell result; a macro automates a sequence of operations.
- What is a relative reference macro? One that records actions relative to the currently selected cell when that option is enabled.
- Why use macros? To reduce repetitive manual work and improve consistency.

---

# Module 4 — Assignment 4: Case studies and project work

These assignments combine the concepts you have already studied. The examiner may ask you to recreate a smaller version of a dashboard or explain how its components work together.

## Assignment 29 — Financial data analysis and budget planner

### What is the task?

Create a workbook that compares actual income and expenditure with the budget, calculates savings and overspending, and displays the results through charts and summary indicators.

### Concepts involved

- `SUM`, `SUMIF`, `IF`, `IFERROR`
- Budget variance and variance percentage

- Conditional formatting
- Charts
- Summary metrics
- Data validation, where appropriate

### Step 1: Create a worksheet

Create a sheet called `Budget Planner` and enter:

| Category | Budget | Actual | Variance | Status |
|---|---|---|---|---|
| Rent | 15000 | 15000 | | |
| Salaries | 30000 | 32000 | | |
| Marketing | 5000 | 4500 | | |
| Travel | 3000 | 4000 | | |
| Utilities | 2000 | 1800 | | |

Enter the headers in row 1 and data in rows 2–6.

### Step 2: Calculate variance

In D2:

```excel
=C2-B2
```

Copy down.

For expenses:

- Positive variance = overspending.
- Negative variance = underspending.
- Zero = exactly on budget.

### Step 3: Calculate variance percentage

If you add a Variance % column in F:

```excel
=IFERROR(D2/B2,0)
```

Format the column as Percentage.

`IFERROR` avoids a division error when the budget is zero. In a financial report, you may prefer a message such as `"No Budget"` instead of zero, because zero could misrepresent the situation.

### Step 4: Classify each category

In E2:

```excel
=IF(C2>B2,"Over Budget",IF(C2<B2,"Under Budget","On Budget"))
```

Copy down.

### Step 5: Calculate total budget and expenditure

Below the table:

```excel
=SUM(B2:B6)
```

This gives total budget.

```excel
=SUM(C2:C6)
```

This gives total actual expenditure.

Overall variance:

```excel
=SUM(C2:C6)-SUM(B2:B6)
```

### Step 6: Highlight overspending

Select the variance cells, D2:D6.

Go to Home → Conditional Formatting → New Rule → Use a formula.

```excel
=D2>0
```

Choose red fill.

For categories under budget, create another rule:

```excel
=D2<0
```

Choose green fill.

### Step 7: Insert a chart

1. Select Category and Actual columns.
2. Go to Insert → Column or Bar Chart.
3. Select Clustered Column.
4. Add a meaningful title, such as `Actual Expenditure by Category`.

You can create a second chart comparing Budget against Actual by selecting Category, Budget, and Actual.

### Step 8: Add summary indicators

At the top of the sheet, show:

- Total Budget
- Total Actual Expenditure
- Overall Variance
- Number of Over-Budget Categories

For the final indicator:

```excel
=COUNTIF(E2:E6,"Over Budget")
```

**Viva questions**

- What is budget variance? The difference between actual and budgeted amounts.
- Why use conditional formatting? To make overspending visible immediately.
- Why use a chart? To compare categories visually rather than inspect every value individually.
- Can positive variance always be considered bad? No. For expenses, positive variance normally means overspending; for income, it can mean exceeding the target.

---

## Assignment 30 — Integrated business case: Employee performance, sales and AI-powered data types

This is the most comprehensive assignment because it combines several Excel concepts into one business workbook.

### What should the workbook contain?

You can organize it into four sheets:

**Sheet 1 — Employee Performance**

Scores, attendance, completion status, classification and bonus eligibility.

**Sheet 2 — Sales and Geography**

City, region, revenue, quantity sold, and geographic information where supported.

**Sheet 3 — Inventory**

Product, category, stock quantity, and low-stock flags.

**Sheet 4 — Executive Dashboard**

Summary metrics, top performers, leading sales regions, and inventory alerts.

### Step 1: Build employee performance calculations

Assume:

- C2 = Performance Score
- D2 = Attendance Percentage
- E2 = Project Status

In F2, classify the employee:

```excel
=IF(AND(C2>=85,D2>=90%,E2="Completed"),"Outstanding",IF(OR(C2<60,D2<75%,E2="Overdue"),"Needs Improvement","Satisfactory"))
```

In G2, calculate bonus eligibility:

```excel
=IF(F2="Outstanding",20%,IF(F2="Satisfactory",10%,0%))
```

If the employee's salary is in H2, the bonus amount can be calculated in I2:

```excel
=H2*G2
```

These percentages are illustrative; use the exact rules in the task.

### Step 2: Build the sales analysis

Suppose city names are in A2:A20, revenue is in B2:B20, and quantity sold is in C2:C20.

Total revenue:

```excel
=SUM(B2:B20)
```

Average revenue:

```excel
=AVERAGE(B2:B20)
```

You can use a PivotTable to summarize revenue by city or region.

For a top-performing city, sort the regional totals from largest to smallest, or apply a Top 10 Value Filter set to the desired number of items.

### Step 3: Add a low-stock indicator

Suppose current stock is in B2 and the reorder threshold is in C2.

```excel
=IF(B2<C2,"Low Stock","Sufficient Stock")
```

You can then use conditional formatting to highlight Low Stock in red.

This is a rule-based inventory alert; it does not predict future demand by itself.

### Step 4: Use Geography data types if available

1. Create a list of valid city names.
2. Select the city cells.
3. Go to Data → Data Types → Geography, if the feature is available.
4. Verify that Excel recognizes the intended locations.
5. Use the available linked fields to enrich the dataset.

If Geography data types are unavailable, you can use a separate reference table of cities and their regions, then retrieve the region using XLOOKUP.

For example:

```excel
=XLOOKUP(A2,$H$2:$H$20,$I$2:$I$20,"Region Not Found")
```

### Step 5: Create the executive dashboard

Create summary metrics such as:

- Total sales revenue
- Best-performing city or region
- Number of low-stock products
- Number of outstanding employees
- Total employee bonuses

For example:

```excel
=COUNTIF(F2:F100,"Outstanding")
```

```excel
=COUNTIF(D2:D100,"Low Stock")
```

The second example assumes your Low Stock labels are in D2:D100 on the inventory sheet. Adjust the range to match your actual layout.

Add a sales chart, a table of top performers, and a low-stock summary. The dashboard should make the most important information visible without requiring the user to inspect every record.

**Viva questions**

- What is a KPI? A Key Performance Indicator used to measure progress toward an objective.
- What is data enrichment? Adding useful information to existing records, such as geographic attributes.
- Why use separate worksheets? To organize related data logically and simplify maintenance.
- Why use a PivotTable? To summarize large transaction datasets efficiently.
- What is the difference between a dashboard and a worksheet? A worksheet can contain raw data or calculations; a dashboard is designed to present key results and insights.

---

# Final module — Concepts that can appear in any practical

These concepts are not separate assignments only. They can be used in many of the tasks above.

## 1. The most important function reference sheet

| Function | Syntax | Purpose |
|---|---|---|
| IF | `=IF(condition,true_result,false_result)` | Conditional result |
| AND | `=AND(condition1,condition2)` | All conditions must be true |
| OR | `=OR(condition1,condition2)` | At least one condition must be true |
| SUM | `=SUM(A2:A10)` | Total |
| AVERAGE | `=AVERAGE(A2:A10)` | Mean |
| AVERAGEIF | `=AVERAGEIF(A2:A10,criteria,B2:B10)` | Conditional average |
| COUNTIF | `=COUNTIF(A2:A10,criteria)` | Conditional count |
| SUMIF | `=SUMIF(A2:A10,criteria,B2:B10)` | Conditional sum |
| IFERROR | `=IFERROR(formula,fallback)` | Handles many errors |
| IFNA | `=IFNA(formula,fallback)` | Handles `#N/A` |
| VLOOKUP | `=VLOOKUP(value,table,column,FALSE)` | Vertical lookup |
| INDEX | `=INDEX(range,position)` | Retrieves a value by position |

| Function | Syntax | Purpose |
|---|---|---|
| MATCH | `=MATCH(value,range,0)` | Finds an exact-match position |
| XLOOKUP | `=XLOOKUP(value,lookup_range,result_range,"Not Found")` | Flexible lookup |
| LEFT | `=LEFT(A2,3)` | Extracts from the left |
| RIGHT | `=RIGHT(A2,3)` | Extracts from the right |
| MID | `=MID(A2,3,5)` | Extracts from a specified position |
| LEN | `=LEN(A2)` | Counts characters |
| FIND | `=FIND("@",A2)` | Finds text, case-sensitively |
| SEARCH | `=SEARCH("@",A2)` | Finds text, case-insensitively |
| SUBSTITUTE | `=SUBSTITUTE(A2,"/","-")` | Replaces matching text |
| TRIM | `=TRIM(A2)` | Cleans excess spaces |
| TODAY | `=TODAY()` | Current date |
| DATE | `=DATE(2026,10,27)` | Creates a date |
| DATEDIF | `=DATEDIF(A2,B2,"y")` | Difference in complete years |
| EOMONTH | `=EOMONTH(A2,0)` | Month-end date |
| RANK.EQ | `=RANK.EQ(A2,$A$2:$A$20,0)` | Descending rank |
| MAX | `=MAX(A2:A20)` | Largest value |
| MAXIFS | `=MAXIFS(B2:B20,A2:A20,"West")` | Maximum matching a criterion |

The argument separators shown are commas. Some Excel regional settings use semicolons instead.

## 2. Know the difference between these pairs

| Pair | Key distinction |
|---|---|
| Conditional formatting vs data validation | Formatting changes appearance; validation controls or restricts input |
| Sorting vs filtering | Sorting reorders records; filtering hides nonmatching records |
| Formula vs macro | A formula calculates a result; a macro automates actions |
| IFERROR vs IFNA | IFERROR handles many error types; IFNA handles only `#N/A` |
| VLOOKUP vs INDEX-MATCH | VLOOKUP searches the first table column; INDEX-MATCH offers more flexible independent ranges |
| PivotTable vs PivotChart | PivotTable summarizes data; PivotChart visualizes a PivotTable |
| Relative vs absolute reference | Relative references change when copied; absolute references remain fixed |
| SUMIF vs COUNTIF | SUMIF totals qualifying values; COUNTIF counts qualifying cells |
| Data cleaning vs Remove Duplicates | Data cleaning includes many corrections; Remove Duplicates removes repeated records based on selected fields |

## 3. How to answer a viva question confidently

For almost any assignment, use this four-part structure:

1. **Purpose:** What problem does the feature solve?
2. **Method:** Which Excel feature or function did you use?
3. **Logic:** Why does the formula or method work?
4. **Result:** What output should the examiner expect?

For example, if asked about your overdue-order formula, you could say:

> *"I used formula-based conditional formatting with AND and a date comparison. AND ensures that the order status is Pending and the due date is before the reference date. The formula highlights the entire matching row, so overdue orders can be identified immediately without manually inspecting every record."*

That answer explains the purpose, function, logic, and result.

## 4. Your final practice checklist

### Exam readiness 0/12

- [ ] Create an above-average conditional-formatting rule.
- [ ] Highlight overdue Pending orders across complete rows.
- [ ] Create a date validation rule and a numeric validation rule.
- [ ] Create a category dropdown and a dependent subcategory dropdown.
- [ ] Write an IF formula containing AND and OR.
- [ ] Write a nested IF formula for grades.
- [ ] Extract an email domain and format a name using text functions.
- [ ] Use IFERROR or IFNA with a lookup formula.
- [ ] Calculate conditional averages and counts using AVERAGEIF and COUNTIF.
- [ ] Create a PivotTable, summarize sales by region, and apply a Top 3 filter.
- [ ] Insert a chart and explain what the chart shows.
- [ ] Explain Freeze Panes, Remove Duplicates, slicers, and macros.

Reset checklist

**My recommendation:** Practise the ten suggested assignments first, especially conditional formatting, dependent dropdowns, IF/AND/OR, nested IF, text functions, IFERROR/IFNA, AVERAGEIF/COUNTIF, and PivotTables. Then use the remaining assignments to prepare for broader viva ques-

tions. For each practical, make sure you can reproduce it from a blank worksheet, not just recognize the finished result.

YOU

now also explain me all the excel concepts used in the assignment theory so I can properly give viva.

# Excel Theory for Practical & Viva Exam

This guide focuses on the **theory behind your Excel assignments**: definitions, working principles, syntax, differences between related features, practical applications, and answers you can give to your examiner.

Your assignments cover four broad areas:

1. Excel essentials, data management, and conditional formatting.
2. Advanced functions, logical formulas, text processing, and data analysis.
3. PivotTables, charts, slicers, and macros.
4. Business case studies, financial analysis, and dashboards.

The objective is to understand *why* each feature is used, not just memorize its formula.

# 1. Excel fundamentals

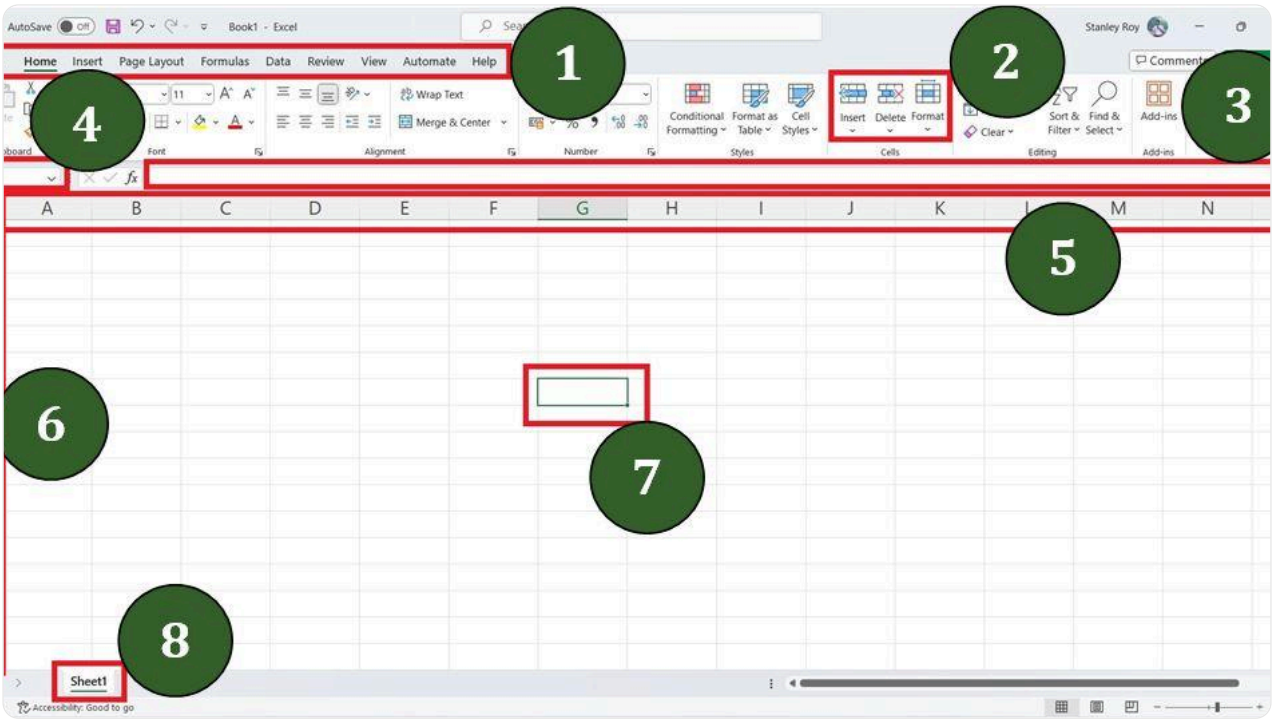
## 1.1 What is Microsoft Excel?

Microsoft Excel is a spreadsheet application used to store, organize, calculate, analyze, and visualize data. It arranges information in rows and columns and provides formulas, functions, charts, PivotTables, and automation tools.

**Applications of Excel:**

- Employee salary and attendance management.
- Sales and inventory analysis.
- Financial budgeting.
- Student result processing.
- Business reporting and dashboards.
- Data cleaning and validation.

## 1.2 Workbook, worksheet, cell, and range




An Excel workbook contains worksheets, and each worksheet contains cells arranged in rows and columns.

| Term | Definition | Example |
|---|---|---|
| Workbook | An Excel file | `Sales.xlsx` |
| Worksheet | An individual sheet in a workbook | Sheet1 |
| Row | Horizontal arrangement of cells | Row 5 |
| Column | Vertical arrangement of cells | Column B |
| Cell | Intersection of a row and column | B5 |
| Cell address | Identifies a cell by column and row | C10 |
| Range | A group of cells | A2:A20 |
| Formula bar | Displays or edits a cell's content or formula | `=SUM(A1:A5)` |
| Ribbon | Contains Excel commands grouped into tabs | Home, Insert, Data |

**Viva answer:** A workbook is the entire Excel file, whereas a worksheet is one spreadsheet inside that file.

## 1.3 Formula versus function

A **formula** is an expression used to calculate a result. A **function** is a predefined operation provided by Excel that can be used in a formula.

Examples:

```excel
=A2+B2
=SUM(A2:A10)
=AVERAGE(B2:B10)
```

The first formula adds two cells. The second and third use predefined functions.

## 1.4 Cell references

| Reference type | Example | Behaviour |
|---|---|---|
| Relative | `A2` | Changes when copied |
| Absolute | `$A$2` | Row and column remain fixed |
| Mixed | `$A2` | Column remains fixed |
| Mixed | `A$2` | Row remains fixed |

For example, if you copy `=B2*$F$1` down one row, it becomes `=B3*$F$1`. The sales reference changes, but the fixed rate in F1 remains the same.

**Why is this important in your assignments?** Conditional formatting, salary calculations, target comparisons, and ranking formulas often require a fixed reference range.

## 1.5 Data types in Excel

Excel can store or interpret:

- Numbers.
- Text.
- Dates and times.
- Logical values (`TRUE` and `FALSE`).
- Formula results.
- Errors such as `#N/A` and `#VALUE!`.

Formatting a cell as a date does not necessarily convert text into a valid date. This matters in overdue-order tracking and deadline monitoring.

---

# 2. Data management and Excel essentials

## 2.1 Sorting

**Definition:** Sorting rearranges data according to one or more fields.

Examples:

- Sales from highest to lowest.
- Names from A to Z.
- Students by program, then city, then admission score.

**How it works:** Select a cell in the dataset and use **Data → Sort**. Add levels for multi-column sorting.

**Viva:** Sorting changes the order of records; it does not remove records.

## 2.2 Filtering

**Definition:** Filtering displays only records that satisfy specified conditions.

Examples:

- Show only pending orders.
- Show students from Pune.
- Show employees earning more than ₹50,000.

Use **Data → Filter**, then select values or number/date filters.

**Difference between sorting and filtering:** Sorting rearranges records, whereas filtering hides nonmatching records temporarily.

## 2.3 Remove Duplicates

**Definition:** Remove Duplicates deletes repeated records based on the columns selected for comparison.

Steps:

1. Select the dataset.
2. Go to Data → Remove Duplicates.
3. Select the columns that define a duplicate.
4. Confirm the operation.

**Important:** Two rows with the same Patient ID may represent legitimate separate hospital visits. The correct duplicate criteria depend on the meaning of the data.

### 2.4 Freeze Panes

**Definition:** Freeze Panes keeps selected rows and columns visible while scrolling.

Steps:

1. Select the cell immediately below the rows and to the right of the columns you want to freeze.
2. Choose **View → Freeze Panes → Freeze Panes**.

For example, selecting C8 freezes rows 1–7 and columns A–B.

**Viva:** Freeze Panes improves navigation in large datasets by keeping headers and identifying columns visible.

### 2.5 Data cleaning

Data cleaning means identifying and correcting inconsistent, invalid, duplicated, or missing data.

Examples:

- Removing unnecessary spaces with `TRIM`.
- Standardizing invoice separators with `SUBSTITUTE`.
- Removing duplicates.
- Correcting inconsistent date formats.
- Validating input values.

**Why is it important?** Incorrect or inconsistent data can lead to wrong calculations, failed lookups, and misleading reports.

---

## 3. Conditional formatting

### 3.1 Definition

Conditional formatting automatically changes a cell's appearance when a specified condition is satisfied.

It can apply:

- Fill colours and font colours.
- Data bars.
- Colour scales.
- Icon sets.
- Formula-based formatting rules.

## 3.2 Types of conditional formatting

| Type | Purpose | Example |
|---|---|---|
| Highlight Cell Rules | Highlights cells meeting conditions | Sales greater than 50,000 |
| Top/Bottom Rules | Highlights top or bottom values | Top 10 sales |
| Data Bars | Shows relative magnitude | Longer bars for higher sales |
| Colour Scales | Shows values through colour gradients | Low-to-high sales |
| Icon Sets | Uses icons to indicate performance | Green, yellow, and red indicators |
| Formula-based rules | Uses a logical formula | Pending orders past their due dates |

## 3.3 How formula-based conditional formatting works

Suppose sales are in B2:B20.

```excel
=B2>AVERAGE($B$2:$B$20)
```

Excel evaluates this formula for each selected cell. If the expression returns TRUE, it applies the specified formatting.

For highlighting entire rows based on the sales value in column B:

```excel
=$B2>AVERAGE($B$2:$B$20)
```

The dollar sign before B fixes the column. The row remains relative so that Excel checks each record.

## 3.4 Conditional formatting versus data validation

| Conditional formatting | Data validation |
|---|---|
| Controls appearance | Controls or restricts input |
| Highlights overdue orders | Restricts dates to a valid period |
| Shows red for overspending | Restricts numbers to an allowed range |
| Does not itself change the underlying value | Can reject invalid entries when configured appropriately |

**Viva answer:** Conditional formatting helps identify important values visually, while data validation helps prevent incorrect data entry.

---

# 4. Data validation and dependent dropdowns

## 4.1 What is data validation?

Data validation is an Excel feature that controls which values users can enter into a cell.

For example, an HR sheet may allow leave dates only within a financial year, or marks only between 0 and 100.

### Types of data validation

| Validation type | Use |
|---|---|
| Whole number | Restrict entries to integers |
| Decimal | Allow numbers within a range |
| List | Create a dropdown |
| Date | Restrict date entries |
| Time | Restrict time entries |
| Text length | Limit the number of characters |
| Custom | Use a formula to test validity |

### Example: Marks between 0 and 100

1. Select the marks cells.
2. Go to Data → Data Validation.
3. Select Allow: Whole number.
4. Select Between.
5. Enter Minimum `0` and Maximum `100`.
6. Configure the error alert if required.

### Example: Dates within a financial year

Choose Allow: Date, set Data to Between, and enter the first and last permissible dates.

**Viva question: What happens if invalid data is entered?**

Depending on the error-alert settings, Excel can reject the entry, warn the user, or allow it after confirmation.

## 4.2 What is a dropdown list?

A dropdown list allows a user to select a value from predefined options rather than typing it manually.

Example: Leave Type may offer Casual Leave, Sick Leave, and Earned Leave.

To create one:

1. Enter the options in a source range.
2. Select the input cells.
3. Go to Data → Data Validation.
4. Choose List.
5. Select the source range.

## 4.3 What is a dependent dropdown?

A dependent dropdown changes its available choices according to the value selected in another dropdown.

Example:

- Category = Electronics → Laptop, Mobile, Monitor.
- Category = Stationery → Pen, Notebook, File.
- Category = Furniture → Chair, Table, Desk.

One common method uses named ranges and `INDIRECT`.

```excel
=INDIRECT(A2)
```

If A2 contains `Electronics`, Excel interprets the text as a reference to a named range called Electronics.

**Advantages:**

- Reduces incorrect entries.
- Makes forms easier to use.
- Maintains consistency in the dataset.

**Limitation:** The selected category must correspond to a valid named range. Category names containing spaces need suitable handling.

### 4.4 What is a named range?

A named range is a cell or group of cells assigned a meaningful name.

For example, instead of referring to `$B$2:$B$10`, you could name the range `SalesData`.

Named ranges improve readability and can simplify formulas and dependent dropdowns.

---

## 5. Logical functions: IF, AND, OR, and nested IF

These functions are among the most important in your advanced-functions assignment.

### 5.1 IF function

**Definition:** IF evaluates a condition and returns one result if it is TRUE and another if it is FALSE.

Syntax:

```excel
=IF(logical_test,value_if_true,value_if_false)
```

Example:

```excel
=IF(B2>=40,"Pass","Fail")
```

If B2 is 65, Excel returns Pass. If B2 is 30, it returns Fail.

### 5.2 AND function

**Definition:** AND returns TRUE only if all its conditions are TRUE.

```excel
=AND(B2>=90,C2>=95%)
```

This returns TRUE only when the score is at least 90 and attendance is at least 95%.

### 5.3 OR function

**Definition:** OR returns TRUE if at least one condition is TRUE.

```excel
=OR(B2<40,C2<75%)
```

This returns TRUE if the score is below 40 or attendance is below 75%, or both.

## 5.4 Nested IF

**Definition:** A nested IF is an IF function placed inside another IF function to evaluate multiple outcomes.

Example:

```excel
=IF(B2>=80,"A",IF(B2>=60,"B",IF(B2>=40,"C","Fail")))
```

Excel checks the conditions in order:

- 80 or above → A.
- Otherwise, 60 or above → B.
- Otherwise, 40 or above → C.
- Otherwise → Fail.

The order is essential. The highest threshold must be tested first in this example.

## 5.5 Combining IF, AND and OR

Example employee classification:

```excel
=IF(AND(B2>=90,C2>=95%),"Outstanding",
IF(OR(B2<60,C2<75%),"Needs Improvement","Satisfactory"))
```

This checks the Outstanding conditions first. If they are not satisfied, it checks whether either critical deficiency exists.

**Viva questions**

- What is a logical test? An expression that evaluates to TRUE or FALSE.
- Why combine logical functions? To implement business rules involving multiple conditions.
- What happens if multiple IF conditions could be true? Excel returns the result for the first satisfied condition.
- What is an alternative to nested IF? `IFS` can handle multiple ordered conditions, while lookup tables can make large classification systems easier to maintain.

---

# 6. Statistical and conditional functions

These functions are essential for employee salaries, attendance, sales, and financial analysis.

## 6.1 SUM

Adds numeric values.

```excel
=SUM(B2:B10)
```

## 6.2 AVERAGE

Calculates the arithmetic mean of numeric values.

```excel
=AVERAGE(B2:B10)
```

For values 10, 20, and 30, the average is 20.

## 6.3 COUNT, COUNTA, and COUNTBLANK

- `COUNT` counts numeric cells.
- `COUNTA` counts non-empty cells.
- `COUNTBLANK` counts blank cells.

Examples:

```excel
=COUNT(A2:A20)
=COUNTA(A2:A20)
=COUNTBLANK(A2:A20)
```

## 6.4 COUNTIF

Counts cells that satisfy one criterion.

Syntax:

```excel
=COUNTIF(range,criteria)
```

Example:

```excel
=COUNTIF(C2:C100,">50000")
```

This counts salaries greater than 50,000.

For a threshold stored in F2:

```excel
=COUNTIF(C2:C100,">"&F2)
```

The `&` joins the comparison operator with the threshold value.

## 6.5 AVERAGEIF

Calculates the average of values meeting a condition.

Syntax:

```excel
=AVERAGEIF(criteria_range,criteria,average_range)
```

Example:

```excel
=AVERAGEIF(B2:B100,"IT",C2:C100)
```

If B contains department names and C contains salaries, this calculates the average salary for IT employees.

## 6.6 SUMIF

Adds values meeting a condition.

```excel
=SUMIF(B2:B100,"IT",C2:C100)
```

This calculates the total salary paid to IT employees.

## 6.7 MAX and MIN

- `MAX` returns the largest value.
- `MIN` returns the smallest value.

```excel
=MAX(C2:C100)
=MIN(C2:C100)
```

These are useful for highest sales, lowest marks, and checking minimum subject scores.

## 6.8 RANK.EQ

Returns the rank of a value within a range.

```excel
=RANK.EQ(E2,$E$2:$E$20,0)
```

The final `0` ranks values in descending order, so the highest score receives rank 1.

**Viva:** COUNTIF counts qualifying cells; SUMIF adds qualifying values; AVERAGEIF averages qualifying values.

---

# 7. Lookup functions

Lookup functions retrieve information from another range or table using a matching value.

## 7.1 VLOOKUP

VLOOKUP searches vertically in the first column of a table and returns a value from a specified column in the matching row.

Syntax:

```excel
=VLOOKUP(lookup_value,table_array,col_index_num,range_lookup)
```

Example:

```excel
=VLOOKUP(E2,A2:C100,2,FALSE)
```

Meaning:

- `E2`: value to search for.
- `A2:C100`: lookup table.
- `2`: return the second column of that table.
- `FALSE`: require an exact match.

### Exact versus approximate match

- `FALSE`: exact match.
- `TRUE`: approximate match, typically used for ordered thresholds such as sales slabs.

For approximate matching, the first column must be sorted in ascending order.

## 7.2 INDEX

Returns a value at a specified position in a range.

```excel
=INDEX(B2:B10,3)
```

Returns the third value in the range B2:B10.

## 7.3 MATCH

Returns the relative position of a value within a range.

```excel
=MATCH(E2,A2:A100,0)
```

The `0` requests an exact match.

### 7.4 INDEX-MATCH

Combines MATCH to find a position and INDEX to return the corresponding value.

```excel
=INDEX(B2:B100,MATCH(E2,A2:A100,0))
```

This is useful for retrieving product names, suppliers, prices, and student details.

### 7.5 XLOOKUP

A modern lookup function that searches one range and returns a corresponding value from another.

```excel
=XLOOKUP(E2,A2:A100,B2:B100,"Not Found")
```

Unlike VLOOKUP, it does not require the lookup column to be the first column of a rectangular table.

### 7.6 IFERROR and IFNA

These functions handle errors produced by other formulas.

```excel
=IFERROR(VLOOKUP(E2,A2:C100,2,FALSE),"Not Found")
```

```excel
=IFNA(VLOOKUP(E2,A2:C100,2,FALSE),"Not Found")
```

- IFERROR handles many error types.
- IFNA handles only `#N/A`.

**Viva question: Why use error handling?**

To provide meaningful messages when a lookup fails and make a spreadsheet easier to use. IFNA is often preferable when you specifically expect a missing match, because it leaves unrelated errors visible.

---

## 8. Text functions

Text functions extract, combine, count, or standardize text. Your assignments use them for email domains, names, member IDs, and invoice numbers.

### 8.1 LEFT

Extracts characters from the beginning of a text string.

```excel
=LEFT(A2,3)
```

If A2 contains `INVOICE`, the result is `INV`.

## 8.2 RIGHT

Extracts characters from the end.

```excel
=RIGHT(A2,3)
```

For `INVOICE`, the result is `ICE`.

## 8.3 MID

Extracts characters from a specified starting position.

```excel
=MID(A2,3,4)
```

For `INVOICE`, this returns `VOIC`.

Syntax: `MID(text,start_num,num_chars)`.

## 8.4 LEN

Counts characters, including spaces and punctuation.

```excel
=LEN(A2)
```

For `LIB-1001`, the result is 8.

## 8.5 FIND and SEARCH

Both return the position of text inside another text string.

```excel
=FIND("@",A2)
=SEARCH("@",A2)
```

Key difference:

- FIND is case-sensitive.
- SEARCH is not case-sensitive.

## 8.6 Extract an email domain

Suppose A2 contains `student@example.com`.

```excel
=RIGHT(A2,LEN(A2)-SEARCH("@",A2))
```

Result: `example.com`.

The formula counts the characters after the @ symbol and extracts that many characters from the right.

## 8.7 UPPER, LOWER and PROPER

- `UPPER` converts text to uppercase.
- `LOWER` converts text to lowercase.
- `PROPER` capitalizes the first letter of each word.

```excel
=UPPER(A2)
=LOWER(A2)
=PROPER(A2)
```

For `krishna patil`, PROPER returns `Krishna Patil`.

## 8.8 SUBSTITUTE

Replaces matching text with other text.

```excel
=SUBSTITUTE(A2,"/","-")
```

For `BILL/26/MUM/102`, this returns `BILL-26-MUM-102`.

## 8.9 TRIM

Removes leading and trailing spaces and reduces repeated internal spaces to a single space.

```excel
=TRIM(A2)
```

Useful for cleaning names and correcting inconsistent text entries.

## 8.10 Concatenation using &

Combines text values.

```excel
=A2&" "&B2
```

This joins two cells with a space between them.

For a name in `Firstname Lastname` format, a two-part name can be converted to `LASTNAME, Firstname` using:

```excel
=UPPER(RIGHT(A2,LEN(A2)-SEARCH(" ",A2)))&", "&LEFT(A2,SEARCH(" ",A2)-1)
```

This assumes the name has exactly two parts.

**Viva questions**

- Why are text functions useful? They standardize inconsistent records and extract information embedded in strings.
- What does LEN count? Every character, including spaces.
- What is the difference between FIND and SEARCH? FIND is case-sensitive; SEARCH is not.
- Why use SUBSTITUTE? To replace specific text or delimiters without manually editing every record.

---

# 9. Date and time functions

Your assignments use dates to track overdue orders, project deadlines, and employee service tenure.

## 9.1 TODAY

Returns the current date.

```excel
=TODAY()
```

It updates when Excel recalculates. It does not insert a permanently fixed date.

## 9.2 DATE

Creates a valid Excel date from year, month, and day values.

```excel
=DATE(2026,10,27)
```

This is useful when constructing dates in formulas.

## 9.3 Date subtraction

Excel stores dates as serial numbers, which allows date arithmetic.

```excel
=B2-A2
```

If A2 is a start date and B2 is an end date, this returns the number of days between them.

For inclusive leave duration:

```excel
=B2-A2+1
```

The `+1` includes both the starting day and the ending day.

## 9.4 DATEDIF

Calculates the difference between two dates in a specified unit.

```excel
=DATEDIF(A2,B2,"y")
```

Returns complete years between the dates.

Common units include:

- `"y"` — complete years.
- `"m"` — complete months.
- `"d"` — days.
- `"ym"` — remaining months after complete years.

The start date must not be later than the end date.

## 9.5 EOMONTH

Returns the last day of a month, offset by a specified number of months.

```excel
=EOMONTH(A2,0)
```

Returns the end of the month containing the date in A2.

```excel
=EOMONTH(A2,1)
```

Returns the end of the following month.

## 9.6 Overdue and deadline calculations

Days remaining:

```excel
=B2-TODAY()
```

Flag a deadline from today through seven days ahead:

```excel
=AND(B2>=TODAY(),B2<=TODAY()+7)
```

Highlight a Pending order whose due date has passed:

```excel
=AND($C2="Pending",$E2<TODAY())
```

**Viva:** Why use TODAY instead of a manually entered current date? The formula updates automatically, whereas a fixed date must be changed manually.

---

# 10. PivotTables and data summarization

## 10.1 What is a PivotTable?

A PivotTable is an interactive Excel tool for summarizing and analyzing a dataset by grouping records and calculating aggregates.

For example, a sales dataset with thousands of transactions can be summarized into total sales by region, category, or salesperson without manually writing a SUMIF formula for every group.

## 10.2 The four PivotTable areas

| Area | Purpose | Example |
|---|---|---|
| Rows | Groups data vertically | Region |
| Columns | Groups data horizontally | Product Category |
| Values | Calculates summaries | Sum of Sales |
| Filters | Restricts the displayed report | Sales Year |

Example:

- Rows → Region
- Columns → Product Category
- Values → Sum of Sales Amount
- Filters → Year

This produces a cross-tabulation of sales by region and product category.

## 10.3 Value Field Settings

Excel can summarize numeric data in several ways:

- Sum
- Count
- Average
- Maximum

- Minimum

If the assignment asks for total sales, use Sum. If it asks for the average package, use Average.

If Excel shows Count instead of Sum, check whether the source values are numeric or stored as text.

## 10.4 Refreshing a PivotTable

When source data changes, the PivotTable may need to be refreshed.

Use:

**Right-click PivotTable → Refresh**

Or select Data → Refresh All.

When new rows are added, a source Excel Table can help the source range expand automatically, although the PivotTable still needs refreshing.

## 10.5 Top 3 filtering

A Top 3 filter identifies the three items with the highest aggregated values.

For regional sales:

1. Insert a PivotTable.
2. Put Region in Rows.
3. Put Sales Amount in Values as Sum.
4. Open the Region dropdown.
5. Choose Value Filters → Top 10.
6. Set it to Top 3 Items by Sum of Sales Amount.

This is different from a label filter: a label filter selects named regions, while a value filter selects regions according to their calculated totals.

**Viva:** Why are PivotTables useful for large datasets? They summarize repeated categories efficiently, allow interactive filtering, and make comparisons easier.

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# 11. Charts, PivotCharts and slicers

## 11.1 What is a chart?

A chart is a visual representation of data that makes comparisons, trends, and patterns easier to understand.

| Chart type | Best use | Example from your assignments |
|---|---|---|
| Column chart | Compare categories | Sales by region |
| Bar chart | Compare categories with long names | Sales by branch |
| Line chart | Show changes over time | Monthly category sales |
| Pie chart | Show parts of a whole | Expense distribution |
| Scatter chart | Examine relationships between two numeric variables | Sales versus advertising spend |

### How to insert a chart

1. Select the data and its headings.
2. Go to Insert.
3. Choose the appropriate chart type.
4. Add a chart title and labels where useful.
5. Check that the chart represents the intended data correctly.

**Viva:** A chart makes data easier to interpret visually. The chart type should match the question being analyzed.

## 11.2 What is a PivotChart?

A PivotChart is a chart linked to a PivotTable. It visualizes summarized data and can respond to PivotTable filtering.

For example, a PivotTable can calculate total monthly sales by product category, and a PivotChart can show each category's trend over six months.

**Difference:** A standard chart can be based directly on a data range; a PivotChart is linked to Pivot-Table-style summarized data.

## 11.3 What is a slicer?

A slicer is a visual filtering control that displays buttons for available categories.

For example, a sales dashboard may have slicers for Region and Product Category. Clicking West filters the report to the West region.

To create one:

1. Select a PivotTable.
2. Choose PivotTable Analyze → Insert Slicer.
3. Select the desired fields.
4. Click OK.
5. Test the slicer buttons.

Slicers can connect to multiple PivotTables when those PivotTables share a compatible data source or data model and the report connections are configured appropriately.

**Viva:** A slicer is more visual and convenient than repeatedly opening filter dropdowns.

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# 12. Macros and VBA

## 12.1 What is a macro?

A macro is a recorded or programmed sequence of actions that automates repetitive tasks.

In your monthly sales report assignment, a macro might:

- Format headings.
- Apply number formats.
- Add totals.
- Apply conditional formatting.
- Create or format a summary sheet.

## 12.2 What is VBA?

VBA stands for Visual Basic for Applications. It is the programming language used to create and edit macros in desktop Microsoft Office applications.

## 12.3 How macro recording works

1. Enable the Developer tab if needed.
2. Go to Developer → Record Macro.
3. Enter a macro name.

4. Perform the actions to record.
5. Click Stop Recording.
6. Run the macro to repeat the recorded actions.

The recorded macro captures a sequence of operations. It does not necessarily understand the business meaning of the data.

### 12.4 Macro-enabled file format

- `.xlsx` — standard Excel workbook; does not store VBA macro code.
- `.xlsm` — macro-enabled Excel workbook; can store VBA macro code.

**Viva questions**

- Why use macros? To automate repetitive tasks, save time, and improve consistency.
- Does a macro automatically adapt to any dataset? Not necessarily. It may depend on fixed ranges or the worksheet layout.
- Is a macro the same as a formula? No. A formula calculates a cell result; a macro performs a sequence of operations.

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## 13. Financial analysis and dashboards

These concepts combine several Excel features from your final assignment.

### 13.1 Budget variance

Variance compares actual results against budgeted or targeted results.

For an expense:

$$\text{Variance} = \text{Actual Expense} - \text{Budget}$$

Example:

- Budget = ₹10,000.
- Actual expense = ₹12,000.
- Variance = ₹2,000 overspending.

In Excel:

```excel
=C2-B2
```

Here B2 contains the budget and C2 the actual expenditure.

For an income target, the interpretation may be reversed: actual income above target is generally favorable.

## 13.2 Variance percentage

$$\text{Variance } \% = \frac{\text{Actual} - \text{Budget}}{\text{Budget}} \times 100$$

Excel formula:

```excel
=IFERROR((C2-B2)/B2,0)
```

This assumes a nonzero budget. If the budget is zero, displaying a meaningful message may be better than displaying zero.

## 13.3 What is a KPI?

KPI stands for **Key Performance Indicator**. It is a measurable value used to evaluate performance against an objective.

Examples from your assignments:

- Total sales revenue.
- Percentage of branches meeting their targets.
- Average employee salary.
- Number of overdue orders.
- Number of low-stock products.
- Average student attendance.

## 13.4 What is a dashboard?

A dashboard is a consolidated view of important metrics, charts, tables, and indicators.

A business dashboard might include:

- Total sales.
- Best-performing region.

- Over-budget categories.
- Employee performance counts.
- Inventory alerts.

**Viva:** A worksheet can contain raw data, calculations, or reports. A dashboard is designed to communicate the most important information quickly.

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# 14. AI-powered data types and data enrichment

Your syllabus includes AI-enhanced data review and a combined business dashboard.

## 14.1 What are data types in this context?

Excel's linked data types can turn supported values, such as recognized geographical names, into records containing additional information.

For example, where supported, a recognized city can be converted into a Geography data type with linked fields.

## 14.2 What is data enrichment?

Data enrichment means adding useful attributes to an existing dataset.

For example, a sales dataset might initially contain only city names. You could enrich it with regional information to analyze sales by region.

## 14.3 How is this useful?

- Adds context to existing records.
- Can reduce manual data entry.
- Supports regional and business analysis.
- Can help identify patterns and anomalies.

Feature availability depends on the Excel version, account, and connectivity. If the linked data-type feature is unavailable, a reference table with XLOOKUP can provide similar enrichment.

**Viva:** AI-powered suggestions and linked data can accelerate analysis, but their outputs should be verified rather than assumed to be correct.

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# 15. Common Excel errors you should recognize

| Error | Meaning | Possible cause |
|---|---|---|
| `#DIV/0!` | Division by zero | Budget or denominator is zero |
| `#N/A` | Value not available | Lookup did not find a match |
| `#VALUE!` | Invalid value type | Formula argument has an incompatible type |
| `#REF!` | Invalid reference | A referenced cell or range was deleted |
| `#NAME?` | Excel does not recognize a name | Misspelled function or undefined name |
| `#NUM!` | Invalid numeric calculation | A number or numeric argument is invalid |
| `#####` | Value cannot be displayed in the current width or format in common cases | Column is too narrow for the displayed value |

For errors caused by a missing lookup, use IFNA or IFERROR where appropriate. Do not hide every error indiscriminately, because some errors reveal mistakes that need to be corrected.

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# 16. Final viva question bank

Use these as oral practice. Try answering each question aloud before revealing the suggested answer.

Theory flashcards **1 of 25**

## What is Excel primarily used for?

**Reveal answer**

**Previous** **Next question**

### How to perform well in the viva

When the examiner asks about a feature, structure your response like this:

1. **Define it.**
2. **Explain its purpose.**
3. **Give an example from your assignment.**
4. **Explain the expected result.**

For example:

> *“Conditional formatting automatically changes the appearance of cells when a specified condition is satisfied. In my sales assignment, I used a formula to compare each branch's sales with the overall average. Excel highlights the branches exceeding that average, making exceptional performance easier to identify.”*

For formula questions, explain what each important argument does. For example, in `=VLOOKUP(E2,A2:C100,2,FALSE)`, explain the lookup value, table range, return-column index, and exact-match argument.

**Final revision priority:** Make sure you can explain conditional formatting, data validation, IF/AND/OR, nested IF, text functions, lookups, conditional statistical functions, date functions, PivotTables, charts, slicers, and macros. These concepts cover the central theory behind your assignment set and give you a strong foundation for follow-up viva questions.